Numerical Computations

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Contents

1 Introduction to Numerical Computations **5**
1.1 Course Information 5
1.2 What Are Numerical Methods? 5
1.2.1 Motivating Example: Roots of Polynomials 6
1.2.2 Numerical Methods in AI, Imaging and Industry 6
1.3 Supercomputers and GPUs 6

2 Errors and Error Estimation **7**
2.1 Why Errors Matter 7
2.1.1 Disasters Caused by Numerical Error 7
2.2 Sources and Types of Error 7
2.3 Accuracy vs. Precision 8
2.4 Absolute and Relative Error 8
2.4.1 Example: Area of a Room 8
2.5 Correct Digits and Rounding 8
2.5.1 Example: Correct Digits of $s = 20.7426$, $\Delta_s = 0.0926$ 8
2.6 Error of Sums and Differences 9
2.6.1 Example: Summing Numbers of Different Accuracy 9
2.6.2 Subtracting Nearly Equal Numbers 9
2.7 Error of Products and Quotients 9
2.7.1 Example: Product/Quotient 10
2.8 Error in Computing Values of Functions 10
2.8.1 Example: $z = \ln(10.3 + \sqrt{4.4})$ 10
2.8.2 Example: Area of a Circle 10
2.9 Inverse Problem: Determining the Error of Arguments 11
2.9.1 Example: Accuracy of an Angle for $\sin x$ with Five Correct Digits 11

3 Computing the Values of Functions **11**
3.1 Overview . . . 11
3.2 Horner's Scheme . . . 11
3.2.1 The Idea . . . 11
3.2.2 Example: $P(-1.5)$ . . . 11
3.2.3 Horner's Scheme as Synthetic Division . . . 12
3.2.4 Example: Dividing by $x - 3$ . . . 12
3.3 Power Series and Taylor's Theorem . . . 12
3.3.1 Evaluating a Function by a Power Series . . . 12
3.3.2 Analytic Functions, Taylor and Maclaurin Series . . . 12
3.3.3 Example: $\sqrt{10}$ by Taylor Series about 9 . . . 13
3.3.4 Example: $\sqrt{10}$ by Maclaurin Series . . . 13
3.4 The Exponential Function . . . 14
3.4.1 A Sharper Remainder Bound . . . 14
3.4.2 Example: $\sqrt{e} = e^{1/2}$ within 10^{-5} . . . 14
3.5 Sine and Cosine . . . 14
3.5.1 Example: $\sin(23°54')$ within 10^{-4} . . . 15
3.6 Hyperbolic Sine and Cosine . . . 15
3.6.1 Example: $\sinh 1.4$ within 10^{-5} . . . 15
3.7 The Natural Logarithm . . . 16
3.7.1 Range Reduction and Error Bound . . . 16
3.7.2 Example: $\ln 3$ within 10^{-5} . . . 16
3.8 Summary of Series Recurrences . . . 16

4 Root-Finding Methods **17**
4.1 Overview . . . 17
4.1.1 Plotting to Find Roots . . . 17
4.1.2 General Structure of an Iterative Method . . . 17
4.1.3 Rate (Degree) of Convergence . . . 17
4.2 Bolzano's Theorem and the IVT . . . 17
4.3 Bisection Method . . . 17
4.3.1 Example: $f(x) = 0.25e^{2x} - 1.5$ . . . 18
4.3.2 Advantages and Disadvantages . . . 18
4.4 False Position (Regula Falsi) Method . . . 18
4.4.1 Example: $2x - \ln x = 7$ . . . 19
4.4.2 Rate of Convergence (Slides' Derivation) . . . 19
4.4.3 Advantages and Disadvantages . . . 19
4.5 Newton–Raphson Method . . . 19
4.5.1 Derivation . . . 19
4.5.2 Quadratic Convergence . . . 20

4.5.3 Example: $x^4 - x = 10$ . . . 20
4.5.4 Example: $e^{-x} = \sin x$ . . . 20
4.5.5 Example: $\sqrt{5}$ (and $\sqrt[k]{a}$ in general) . . . 20
4.5.6 Newton Does Not Always Converge . . . 21
4.5.7 Advantages and Disadvantages . . . 21
4.6 Secant Method . . . 21
4.7 Comparison Example: $\sqrt{11}$ by All Four Methods . . . 21
4.8 Fixed-Point Iteration . . . 22
4.8.1 Convergence Theorem . . . 22
4.8.2 Example: $3x - 2e^{-x} = 0$ . . . 22
4.8.3 Example: Choice of g Matters — $x^2 - x - 2 = 0$ . . . 23
4.8.4 Advantages and Disadvantages . . . 23
4.9 Summary of Root-Finding Methods . . . 23
4.10 Homework Problems (Chapter 2+) . . . 23

5 Systems of Linear Equations 23
5.1 Overview . . . 23
5.2 Notation . . . 24
5.3 Cramer's Rule . . . 24
5.4 The Gaussian Method (Successive Elimination) . . . 24
5.4.1 Warm-up Example (3×3, row operations) . . . 24
5.4.2 The General Scheme (shown for $n = 4$) . . . 24
5.4.3 Operation Count . . . 25
5.4.4 Worked Example (4×4) . . . 25
5.4.5 Check Sums . . . 26
5.5 The Gaussian Compact Scheme . . . 26
5.5.1 Structure of the Table ($n = 4$) . . . 26
5.5.2 Example (same 4×4 system) . . . 26
5.6 Method of Principal Elements (Pivoting) . . . 27
5.6.1 Example: Why Pivoting Matters . . . 27
5.6.2 Example: Principal Elements on the 3×3 System . . . 27
5.7 LU Factorisation: Crout–Doolittle and the Khaletsky Scheme . . . 27
5.7.1 Idea . . . 27
5.7.2 Khaletsky (Crout) Formulas . . . 28
5.7.3 Example: Khaletsky on the 4×4 System . . . 28
5.7.4 Example: Doolittle on the 3×3 System . . . 28
5.8 Iterative Methods: Successive Approximations . . . 28
5.8.1 General Form . . . 28
5.8.2 Jacobi's Method . . . 29
5.8.3 Gauss–Seidel Method . . . 29

5.8.4 Example: Jacobi vs. Gauss–Seidel 29
5.9 Summary of Linear-System Methods 30

6 Systems of Nonlinear Equations 30
6.1 Overview 30
6.2 Newton's Method for Two Equations 30
6.2.1 Derivation 30
6.2.2 Solving the 2×2 System by Cramer's Rule 30
6.2.3 Example 1: $2x^3 - y^2 - 1 = 0, \quad xy^3 - y - 4 = 0$ 31
6.2.4 Example 2: $\cos(0.4y + x^2) + x^2 + y^2 - 1.6 = 0, \ 1.5x^2 - \frac{y^2}{0.36} - 1 = 0$ 31
6.3 Newton's Method for n Equations in n Unknowns 32
6.3.1 Example: $x^2 + y^2 + z^2 = 1, \ 2x^2 + y^2 - 4z = 0, \ 3x^2 - 4y + z^2 = 0$ 32
6.4 Newton as a Fixed-Point Problem 33
6.5 Simple Iteration and the Contraction Mapping Theorem 33
6.5.1 Setting 33
6.5.2 Contraction Mapping 33
6.5.3 A Practical Test via Partial Derivatives 33
6.5.4 Example: Line and Ellipse 34
6.6 Homework (Chapter 4) 34

7 Exam Cheat Sheet (Sections 1–4) 34
7.1 Errors 34
7.2 Values of Functions 35
7.3 Root Finding 35
7.4 Linear Systems 35
7.5 Nonlinear Systems 35

1 Introduction to Numerical Computations

1.1 Course Information

- **Lecturers:** Hamid Sarbazi-Azad (azad@sharif.edu, room 621) and Negar Akbarzadeh (akbarzadeh@ipm.ir, room 703).
- **Head TA:** Atiyeh Gheibi-Fetrat (atiye.gheibi@gmail.com, HPCAN Lab, room 703).
- **Main reference:** N. V. Kopchenova and I. A. Maron, *Computational Mathematics*, Mir Publishers (Persian translation: H. Sarbazi-Azad). GPU references: NVIDIA *CUDA C Programming Guide* (v6.5) and Sanders & Kandrot, *CUDA by Example*.

Grading policy:

Item	Weight	Date
Assignments	10%	
Project	10%	
Midterm (Sections 1–4)	40%	Thursday 26 Azar, 11:00
Final (Sections 5–8)	40%	Thursday 7 Bahman, 15:00

Course outline:

0. Introduction to NC and parallel programming with GPUs
1. Approximate computations and error estimation

2+. Computing the values of functions and solving equations (root finding)

3. Numerical solution of systems of linear equations
4. Numerical solution of systems of nonlinear equations
5. Interpolation of functions
6. Numerical differentiation
7. Numerical integration
8. Approximate solution of ordinary differential equations
9. Boundary value problems for ODEs
10. Numerical solution of PDEs and integral equations

▶ *This document covers Sections 0–4, i.e. everything on the midterm.*

1.2 What Are Numerical Methods?

Definition. Numerical computation is the study of *approximation techniques* for solving mathematical problems numerically.

Very few mathematical problems have closed-form solutions, so numerical methods are necessary. Typical examples:

- extracting the roots of a polynomial of degree > 4;
- computing the eigenvalues of a matrix (e.g. PageRank);
- solving most ODEs and PDEs.

More than 99.9% of real-world science and engineering problems can only be solved numerically. Analytic solutions are rare, but they remain the "core truth": they reveal the structure of a problem and let us test the integrity of numerical schemes.

1.2.1 Motivating Example: Roots of Polynomials

Bisection idea (developed fully in Section 4) for a root of f in $[a, b]$:

1. Pick a, b such that $f(a)$ and $f(b)$ have opposite signs.
2. Bisect (a, b) into (a, m) and (m, b), $m = \frac{a+b}{2}$; keep the half on whose ends f still has opposite signs.
3. Repeat until the interval is short enough; its midpoint is the numerical solution.

Equation	Analytic solution	Numerical (bisection)
$z^3 + a_2 z^2 + a_1 z + a_0 = 0$	Cardano's formula (long)	applies
$z^4 + a_3 z^3 + \cdots + a_0 = 0$	Ferrari's formula (very long)	applies, nothing changes
$z^5 + a_4 z^4 + \cdots + a_0 = 0$	does not exist (Abel–Ruffini), except special cases	applies as usual

▶ *From degree 5 on, numerical methods are the* only *general option.*

1.2.2 Numerical Methods in AI, Imaging and Industry

Local search in continuous spaces (AIMA, 3rd ed.). Place 3 airports so that the sum of squared distances from each city to its nearest airport is minimal. The state is $(x_1, y_1, x_2, y_2, x_3, y_3)$ and

$$f(x_1, x_2, x_3, y_1, y_2, y_3) = \sum_{i=1}^{3} \sum_{c \in C_i} \left[(x_i - x_c)^2 + (y_i - y_c)^2 \right],$$

where C_i is the set of cities closest to airport i. To minimise f we solve $\nabla f = 0$ with

$$\nabla f = \left(\frac{\partial f}{\partial x_1}, \frac{\partial f}{\partial x_2}, \frac{\partial f}{\partial x_3}, \frac{\partial f}{\partial y_1}, \frac{\partial f}{\partial y_2}, \frac{\partial f}{\partial y_3} \right).$$

The most effective algorithm is the **Newton–Raphson** update

$$\mathbf{x} \leftarrow \mathbf{x} - H_f^{-1}(\mathbf{x})\, \nabla f(\mathbf{x}),$$

where H_f is the Hessian. (This is exactly Newton's method for the nonlinear system $\nabla f = 0$; see Section 6.)

Other applications:

- **Image processing:** eigenvalues are used for measuring image sharpness, human-face segmentation and whitening.
- **Google PageRank:** an eigenvector of a matrix of order ~ 3 billion — the world's largest matrix computation.
- **Airlines:** optimisation for ticket prices, aircraft, fuel and crew assignment.
- **Car crash simulation:** numerically solving PDEs.
- **Hedge funds:** numerical analysis to value stocks.

1.3 Supercomputers and GPUs

- Moore's law and parallel architectures have driven supercomputer growth for seven decades.
- Supercomputers are ranked every six months at `top500.org` by computing power in **FLOPS** (floating-point operations per second). Adding or multiplying two IEEE-754 floating-point numbers counts as one FLOP.
- **GPUs** dominate today's massively parallel processing and have been the major enabler of deep neural networks.

Top-10 list (2026, from the slides):

Rank	System (site, country)	Cores	Rmax (PF/s)	Power (kW)
1	LineShine (NSC Shenzhen, China)	13,789,440	2198.40	42,220
2	El Capitan (LLNL, USA)	11,340,000	1809.00	29,685
3	Frontier (ORNL, USA)	9,066,176	1353.00	24,607
4	Aurora (Argonne, USA)	9,264,128	1012.00	38,698
5	JUPITER Booster (FZJ, Germany)	4,801,344	1000.00	15,794
6	HPC7 (Eni, Italy)	3,461,472	571.50	8,735
7	Eagle (Microsoft Azure, USA)	2,073,600	561.20	–
8	HPC6 (Eni, Italy)	3,143,520	477.90	8,461
9	Fugaku (RIKEN, Japan)	7,630,848	442.01	29,899
10	Alps (CSCS, Switzerland)	2,121,600	434.90	7,124

▶ *Today's computer engineers must know parallel programming, especially GPU programming.*

2 Errors and Error Estimation

2.1 Why Errors Matter

Numbers such as $\pi = 3.14159265\ldots$, $e = 2.71828182\ldots$, $\sqrt{5} = 2.2360679\ldots$ and $\frac{1}{3} = 0.3333333\ldots$ can only ever be stored approximately. Numerical methods give *approximate* results, so we must ask: *how much error is present, and is it tolerable?*

We measure errors

- to determine the accuracy of numerical results, and
- to develop **stopping criteria** for iterative algorithms.

2.1.1 Disasters Caused by Numerical Error

Patriot missile (25 Feb 1991, Dhahran). The system failed to intercept a Scud; 28 soldiers were killed. The clock tick of $\frac{1}{10}$ s was stored in a 24-bit fixed-point register. Since 0.1 has no finite binary expansion, each tick carried an error of 9.5×10^{-8} s. After 100 hours of operation:

$$9.5 \times 10^{-8}\ \frac{\text{s}}{\text{tick}} \times 100\,\text{h} \times \frac{3600\,\text{s}}{1\,\text{h}} \times \frac{1\ \text{tick}}{0.1\,\text{s}} = 0.342\,\text{s},$$

enough for a Scud to travel about half a kilometre.

Ariane 5 (4 June 1996). 36 seconds after launch the rocket veered off and self-destructed. The Inertial Reference System raised an operand exception converting a 64-bit float to a small signed integer (the slides say 12-bit; the inquiry report says 16-bit) that was too small to hold it; the diagnostic word was interpreted as flight data. Ironically, the code was legacy from Ariane 4 and its result was not needed after lift-off.

Vancouver Stock Exchange (1982). A new index started at 1000 and was updated after every transaction. After 22 months it had fallen to 520 because updates were *truncated* instead of *rounded*. Correct rounding gave 1098.892.

2.2 Sources and Types of Error

Sources of error:

- **Measurement:** measured data contain error.
- **Representation:** finite representation of numbers in the machine.
- **Mathematical model:** some parameters are ignored when modelling.

Types of error:

- **Round-off error:** computers work with a finite representation, e.g. $\frac{1}{3} \approx 0.33333$, $\sqrt{2} \approx 1.4142$.
- **Truncation error:** an infinite process is cut off, e.g. approximating a function by a polynomial. The Taylor series of e^x about 0,

  $$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots,$$

  is truncated after finitely many terms.

2.3 Accuracy vs. Precision

- **Accuracy:** how close the value is to the *true* value.
- **Precision:** how close repeated measured values are to *each other*.

▶ *Think of a target: shots clustered tightly but off-centre are precise but inaccurate; shots scattered around the bull's-eye are accurate on average but imprecise. The goal is both.*

2.4 Absolute and Relative Error

Let a be an approximation of the exact value a_0.

Absolute error:

$$\Delta_a = |a - a_0| \qquad (\text{always } \geq 0;\ \text{in practice we use an upper bound}).$$

Relative error:

$$\delta_a = \frac{\Delta_a}{|a|} \qquad (a \neq 0).$$

In applied sciences accuracy is usually characterised by the *relative* error:

	Case 1	Case 2
Actual size	17.2 cm	82.49 m
Measured size	18.2 cm	82.50 m
Absolute error	1 cm	1 cm
Relative error	0.05	0.00012

Same absolute error, but the second measurement is far more accurate.

2.4.1 Example: Area of a Room

Length and width measured to 1 cm: $a = 5.43\,\text{m}$, $b = 3.82\,\text{m}$, so $S = ab = 20.7426\,\text{m}^2$.

With $\Delta_a = \Delta_b = 0.01\,\text{m}$, the extreme possible areas are

$$(a + 0.01)(b + 0.01) = 20.8352, \qquad (a - 0.01)(b - 0.01) = 20.6502.$$

Hence $|S - S_0| \leq 0.0926$ and

$$\delta_S = \frac{0.0926}{20.7426} \approx 0.0045 = 0.45\%.$$

▶ *Check with the product rule of Section 2.7:* $\delta_S \approx \delta_a + \delta_b = \frac{0.01}{5.43} + \frac{0.01}{3.82} = 0.00184 + 0.00262 = 0.0045$. *Same answer, much faster.*

2.5 Correct Digits and Rounding

Definition. The **correct digits** of a number are counted from its first significant digit up to the first significant digit (decimal place) of its absolute error.

Rounding rule: if the first discarded digit is ≥ 5, add 1 to the last retained digit.

▶ *When you round a result, the rounding error (the discarded part) must be* ***added*** *to the absolute error.*

2.5.1 Example: Correct Digits of $s = 20.7426$, $\Delta_s = 0.0926$

The first significant digit of Δ_s is in the 10^{-1} place, so the digits $2, 0, 7$ are correct and we write $s = 20.7$. The discarded quantity is $20.7426 - 20.7 = 0.0426$, so

$$\Delta_s = 0.0926 + 0.0426 = 0.1352 \approx 0.136.$$

Now the error's first significant digit is again in the 10^{-1} place, but $0.136 > 0.1$, so strictly the tenths digit is no longer guaranteed. Rounding again to correct digits gives $s = 21$ with

$$\Delta_s = 0.136 + 0.3 = 0.436.$$

▶ *Lesson: rounding a result off to its correct digits is* not always advisable*; each rounding increases the error bound. Keep one or two spare digits during computation.*

2.6 Error of Sums and Differences

Rule. The absolute error of an algebraic sum is at most the sum of the absolute errors:

$$S = a_1 \pm a_2 \pm \cdots \pm a_n \implies \Delta_S = \Delta_{a_1} + \Delta_{a_2} + \cdots + \Delta_{a_n}.$$

If one term's error dominates the others, the error of the sum is taken to be that greatest error.

Remark (same-sign terms). If $a_1, \ldots, a_n$ all have the same sign, then

$$\delta_{\min} \le \delta_S \le \delta_{\max}.$$

Proof sketch (HW#1). $\delta_S = \frac{\sum \Delta_{a_i}}{\sum |a_i|} = \frac{\sum \delta_{a_i} |a_i|}{\sum |a_i|}$ is a weighted average of the δ_{a_i} with positive weights, hence lies between their minimum and maximum. ■

2.6.1 Example: Summing Numbers of Different Accuracy

Sum 0.348, 0.1834, 345.4, 235.2, 11.75, 9.27, 0.0849, 0.0214, 0.000354, all digits correct (each error $\le$ half a unit of the last digit).

The least accurate terms are 345.4 and 235.2 ($\Delta = 0.05$ each). Round all other terms to one spare digit (hundredths) and add:

$$345.4 + 235.2 + 11.75 + 9.27 + 0.35 + 0.18 + 0.08 + 0.02 + 0.00 = 602.25 \approx 602.3.$$

Error: the two dominant terms give $2 \times 0.05 = 0.1$; the final rounding adds 0.05:

$$\Delta_S = 0.1 + 0.05 = 0.15.$$

▶ *Procedure: (1) find the least accurate terms; (2) round the others to one more decimal place than those; (3) add; (4) round the result and add up the errors.*

2.6.2 Subtracting Nearly Equal Numbers

For $c = a - b$ with $a, b > 0$, the relative error of the difference can be much larger than those of a and b, especially when $a \approx b$ (**catastrophic cancellation**).

Example. $a^* = 1.137$, $b^* = 1.073$, $\Delta_a = \Delta_b = 0.011$ (so $\delta_a \approx \delta_b \approx 1\%$):

$$c^* = 0.064, \qquad \Delta_c = 0.022, \qquad \delta_c = \frac{0.022}{0.064} \approx 35\%.$$

▶ *Avoid subtracting nearly equal numbers; rewrite formulas when possible, e.g.* $\sqrt{x+1} - \sqrt{x} = \dfrac{1}{\sqrt{x+1} + \sqrt{x}}$.

2.7 Error of Products and Quotients

Proposition (product). If $p = ab$ then $\delta_p = \delta_a + \delta_b$.

Proof.

$$p_{\max} = (a + \Delta_a)(b + \Delta_b) = ab + b\Delta_a + a\Delta_b + \Delta_a\Delta_b, \qquad p_{\min} = (a - \Delta_a)(b - \Delta_b) = ab - b\Delta_a - a\Delta_b + \Delta_a\Delta_b.$$

Since $\Delta_a, \Delta_b \approx 0$, neglect $\Delta_a\Delta_b$: $\Delta_p \le |b|\,\Delta_a + |a|\,\Delta_b$, so

$$\delta_p \le \frac{|b|\,\Delta_a + |a|\,\Delta_b}{|ab|} = \delta_a + \delta_b. \qquad \blacksquare$$

Proposition (quotient). If $p = \dfrac{a}{b}$ then $\delta_p = \delta_a + \delta_b$.

Proof. $p_{\max} = \frac{a+\Delta_a}{b-\Delta_b}$, $p_{\min} = \frac{a-\Delta_a}{b+\Delta_b}$, and

$$p_{\max} - p = \frac{b\Delta_a + a\Delta_b}{b(b - \Delta_b)}, \qquad p - p_{\min} = \frac{b\Delta_a + a\Delta_b}{b(b + \Delta_b)}.$$

For $\Delta_b \to 0$, $\Delta_p \le \dfrac{|b|\,\Delta_a + |a|\,\Delta_b}{b^2}$, so $\delta_p = \dfrac{\Delta_p}{|a/b|} \le \delta_a + \delta_b$. ■

General rule. For

$$r = \frac{a_1 a_2 \cdots a_m}{b_1 b_2 \cdots b_n}: \qquad \delta_r = \delta_{a_1} + \cdots + \delta_{a_m} + \delta_{b_1} + \cdots + \delta_{b_n}, \qquad \Delta_r = |r|\,\delta_r.$$

Statistical estimate. If $m + n > 10$ and all factors have roughly the same relative error δ, errors of different signs partly cancel, and it is more realistic to use

$$\delta_r = \sqrt{3(m+n)}\,\delta.$$

2.7.1 Example: Product/Quotient

Compute $r = \dfrac{3.2 \times 356.7 \times 0.04811}{7.1948 \times 34.56}$ and its error (all digits correct).

The least accurate factor is 3.2 (two significant digits): $\delta = \dfrac{0.05}{3.2} \approx 0.016$. It dominates, so round the other factors to one spare digit (three significant digits):

$$r = \frac{3.2 \times 357 \times 0.0481}{7.19 \times 34.6} = 0.221, \qquad \Delta_r = r\delta_r = 0.221 \times 0.016 = 0.0036.$$

Rounding to correct digits: $r = 0.22$.

▶ *Rule of thumb: the result of $\times/\div$ has about as many correct significant digits as the* least *accurate factor.*

2.8 Error in Computing Values of Functions

One variable. For differentiable $y = f(x)$ and a small error Δ_x (linearisation via the mean value theorem):

$$\Delta_y = |f'(x)|\,\Delta_x, \qquad \delta_y = \frac{|f'(x)|}{|f(x)|}\Delta_x = \Big|\big[\ln f(x)\big]'\Big|\,\Delta_x.$$

Several variables. For $y = f(x_1, \ldots, x_n)$:

$$\Delta_y = \sum_{i=1}^{n} \left|\frac{\partial f}{\partial x_i}\right| \Delta_{x_i}, \qquad \delta_y = \frac{1}{|f|}\sum_{i=1}^{n} \left|\frac{\partial f}{\partial x_i}\right| \Delta_{x_i} = \sum_{i=1}^{n} \left|\frac{\partial \ln f}{\partial x_i}\right| \Delta_{x_i} \quad (f > 0).$$

▶ *The sum, product and quotient rules above are all special cases: e.g. $\ln(ab) = \ln a + \ln b$ gives $\delta_{ab} = \frac{\Delta_a}{a} + \frac{\Delta_b}{b}$.*

Useful special cases:

- Logarithm: $\Delta_{\ln x} = \dfrac{\Delta_x}{x} = \delta_x$ (absolute error of $\ln x$ = relative error of x).
- Power: $\delta_{x^k} = |k|\,\delta_x$ (e.g. $\delta_{\sqrt{x}} = \frac{1}{2}\delta_x$, $\delta_{x^2} = 2\delta_x$).
- Trigonometric:

$$\Delta_{\sin x} = |\cos x|\,\Delta_x \le \Delta_x, \quad \Delta_{\cos x} = |\sin x|\,\Delta_x \le \Delta_x,$$
$$\Delta_{\tan x} = (1 + \tan^2 x)\Delta_x \ge \Delta_x, \quad \Delta_{\cot x} = (1 + \cot^2 x)\Delta_x \ge \Delta_x.$$

2.8.1 Example: $z = \ln(10.3 + \sqrt{4.4})$

All digits of $x = 10.3$, $y = 4.4$ are correct, so $\Delta_x = \Delta_y = 0.05$.

1. $\delta_y = \dfrac{0.05}{4.4} \approx 1.2\%$, so $\delta_{\sqrt{y}} = \frac{1}{2}\delta_y \approx 0.6\%$.
2. $\sqrt{y} = \sqrt{4.4} = 2.10$, $\Delta_{\sqrt{y}} = 2.10 \times 0.006 = 0.013$.
3. $x + \sqrt{y} = 10.3 + 2.10 = 12.4$, $\Delta = 0.05 + 0.013 = 0.063$, $\delta_{x+\sqrt{y}} = \dfrac{0.063}{12.4} \approx 0.5\%$.
4. $z = \ln 12.4 = 2.517$, and $\Delta_z = \delta_{x+\sqrt{y}} \approx 0.005$.

So $z = 2.517 \pm 0.005$, i.e. $z \approx 2.52$.

2.8.2 Example: Area of a Circle

The diameter, measured to within 1 mm, is $d = 0.842\,\text{m}$. Find $S = \pi d^2/4$ and its error.

π can be taken as accurately as we like, so the error comes from d^2:

$$\delta_{d^2} = 2\delta_d = 2 \times \frac{1}{842} \approx 0.24\%, \qquad \delta_S = \delta_\pi + 2\delta_d.$$

Take π with at least four correct digits so that $\delta_\pi \ll \delta_{d^2}$:

$$S = \frac{3.1416}{4} \times 0.842^2 = 0.7854 \times 0.7090 = 0.5568\,\text{m}^2, \qquad \Delta_S = 0.5568 \times 0.0024 = 0.0013.$$

After rounding to $S = 0.557\,\text{m}^2$: $\Delta_S = 0.0013 + 0.0002 = 0.0015 \approx 0.002$.

2.9 Inverse Problem: Determining the Error of Arguments

Given the permissible error Δ_y of the result, how accurately must the arguments be known?

One variable:

$$\Delta_x = \frac{\Delta_y}{|f'(x)|}, \qquad f'(x) \neq 0.$$

Several variables (principle of **equal effects**: each argument contributes the same share of the error):

$$\Delta_{x_i} = \frac{\Delta_y}{n\,|\partial f/\partial x_i|}, \qquad i = 1, \ldots, n.$$

2.9.1 Example: Accuracy of an Angle for $\sin x$ with Five Correct Digits

For $x > 6^\circ$ (first quadrant) we have $\sin x > 0.1$, so five correct digits require

$$\Delta_{\sin x} < 0.5 \times 10^{-5}.$$

Then

$$\Delta_x = \frac{\Delta_{\sin x}}{|\cos x|}.$$

▶ *The slides write $\Delta_x > 2 \times 0.5 \times 10^{-5} = 10^{-5}$. Since $|\cos x| \leq 1$, the safe requirement valid for every first-quadrant angle is $\Delta_x \leq 0.5 \times 10^{-5}$ rad ($\approx 1''$); near $x = 60^\circ$ ($\cos x = \frac{1}{2}$) one may relax it to 10^{-5} rad ($\approx 2''$).*

3 Computing the Values of Functions

3.1 Overview

Computers natively perform only $+, -, \times, \div$. So computing the value of any function means reducing it to a *sequence of elementary arithmetic operations.* Two tools:

- **Horner's scheme** for polynomials (William George Horner, 1786–1837);
- **Taylor / Maclaurin series** for analytic functions (Colin Maclaurin, 1698–1746), with an error bound that tells us how many terms to keep.

3.2 Horner's Scheme

3.2.1 The Idea

Let $P(x) = a_0 x^n + a_1 x^{n-1} + \cdots + a_{n-1} x + a_n$ be a real polynomial of degree n. We want $P(\xi)$. Factor out ξ repeatedly (nested form):

$$P(\xi) = \Big(\cdots \big((a_0 \xi + a_1)\xi + a_2 \big)\xi + \cdots + a_{n-1} \Big)\xi + a_n.$$

Compute successively

$$b_0 = a_0, \qquad b_k = b_{k-1}\,\xi + a_k \quad (k = 1, \ldots, n), \qquad P(\xi) = b_n.$$

Tableau:

	a_0	a_1	a_2	$\cdots$	a_n
ξ		$b_0\xi$	$b_1\xi$	$\cdots$	$b_{n-1}\xi$
	b_0	b_1	b_2	$\cdots$	$b_n = P(\xi)$

Cost: n multiplications and $n - k$ additions, where k is the number of zero coefficients. (Naive evaluation needs about $2n - 1$ multiplications, or $n(n+1)/2$ if powers are recomputed.)

3.2.2 Example: $P(-1.5)$

$P(x) = x^7 - 2x^6 + x^5 - 3x^4 + 4x^3 - x^2 + 6x - 1$, $\xi = -1.5$:

	1	−2	1	−3	4	−1	6	−1
−1.5		−1.5	5.25	−9.375	18.5625	−33.8438	52.2656	−87.3984
	1	−3.5	6.25	−12.375	22.5625	−34.8438	58.2656	**−88.3984**

So $P(-1.5) \approx -88.3984$.

▶ *The slides print the 7th product as -52.2657; the correct value is $(-34.84375)(-1.5) = +52.265625$, which is what gives 58.2656 in the bottom row.*

3.2.3 Horner's Scheme as Synthetic Division

Divide $P(x)$ by $x-\xi$: $P(x) = Q(x)(x-\xi) + \beta_n$ with $Q(x) = \beta_0 x^{n-1} + \beta_1 x^{n-2} + \cdots + \beta_{n-1}$. Expanding,

$$P(x) = \beta_0 x^n + (\beta_1 - \beta_0\xi)x^{n-1} + (\beta_2 - \beta_1\xi)x^{n-2} + \cdots + (\beta_{n-1} - \beta_{n-2}\xi)x + (\beta_n - \beta_{n-1}\xi).$$

Comparing coefficients with $P(x) = a_0 x^n + \cdots + a_n$:

$$\beta_0 = a_0 = b_0, \qquad \beta_k = a_k + \beta_{k-1}\xi = b_k \quad (k = 1, \ldots, n).$$

▶ *The intermediate numbers $b_0, \ldots, b_{n-1}$ of Horner's scheme are exactly the coefficients of the quotient $Q(x)$, and $b_n = P(\xi)$ is the remainder (the Remainder Theorem). No division is needed.*

3.2.4 Example: Dividing by $x - 3$

$P(x) = x^6 + 2x^5 + 3x^4 + 4x^3 + 5x^2 + 6x - 1$:

	1	2	3	4	5	6	−1
3		3	15	54	174	537	1629
	1	5	18	58	179	543	1628

$$P(x) = (x^5 + 5x^4 + 18x^3 + 58x^2 + 179x + 543)(x - 3) + 1628.$$

▶ *Applying Horner again to Q at the same ξ gives $P'(\xi) = Q(\xi)$ — useful later for Newton's method on polynomials.*

3.3 Power Series and Taylor's Theorem

3.3.1 Evaluating a Function by a Power Series

Suppose $f(0)$ and all $f^{(n)}(0)$ are known and we seek $f(x) = a_0 + a_1 x + a_2 x^2 + \cdots$. Differentiating term by term,

$$f'(x) = a_1 + 2a_2 x + \cdots, \quad \ldots, \quad f^{(n)}(x) = n!\, a_n + \cdots,$$

and putting $x = 0$ gives $a_n = \dfrac{f^{(n)}(0)}{n!}$.

3.3.2 Analytic Functions, Taylor and Maclaurin Series

Definition. f is **analytic** at ξ iff its Taylor series converges to $f(x)$ for every x in some neighbourhood $|x - \xi| < R$.

Taylor series:

$$f(x) = f(\xi) + f'(\xi)(x-\xi) + \frac{f''(\xi)}{2!}(x-\xi)^2 + \cdots + \frac{f^{(n)}(\xi)}{n!}(x-\xi)^n + \cdots$$

Maclaurin series ($\xi = 0$):

$$f(x) = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots$$

Taylor polynomial and remainder. Replacing f by

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(\xi)}{k!}(x-\xi)^k$$

gives the error (remainder term) $R_n(x) = f(x) - P_n(x)$, and there is $0 < \theta < 1$ with (Lagrange form)

$$R_n(x) = \frac{f^{(n+1)}\big(\xi + \theta(x-\xi)\big)}{(n+1)!}(x-\xi)^{n+1}.$$

Mean Value Theorem. If f is continuous on $[a, b]$ and differentiable on (a, b), there is $c \in (a, b)$ with

$$\frac{f(b) - f(a)}{b - a} = f'(c).$$

Taylor's Theorem. Let f be real on $[a,b]$, n a positive integer, $f', \dots, f^{(n)}$ continuous on $[a,b]$ and $f^{(n+1)}$ existing on (a,b). Let $\xi \in (a,b)$. Then for every $x \in (a,b)$ there is c between ξ and x with

$$f(x) = P_n(x) + \frac{f^{(n+1)}(c)}{(n+1)!}(x-\xi)^{n+1}.$$

For $n = 0$ this is the Mean Value Theorem: $f(x) - f(\xi) = f'(c)(x-\xi)$. The theorem lets us *bound the error* once we have a bound on $\left|f^{(n+1)}\right|$.

Sketch of proof. Fix x and choose M so that $f(x) = P_n(x) + M(x-\xi)^{n+1}$. Define

$$g(t) = f(t) - f(\xi) - \frac{f'(\xi)}{1!}(t-\xi) - \cdots - \frac{f^{(n)}(\xi)}{n!}(t-\xi)^n - M(t-\xi)^{n+1}.$$

With $a_i = f^{(i)}(\xi)/i!$,

$$g^{(k)}(t) = f^{(k)}(t) - \sum_{i \geq k} \frac{i!}{(i-k)!} a_i (t-\xi)^{i-k} - \frac{(n+1)!}{(n+1-k)!} M(t-\xi)^{n+1-k}, \qquad g^{(n+1)}(t) = f^{(n+1)}(t) - (n+1)!\, M.$$

So $g(\xi) = g'(\xi) = \cdots = g^{(n)}(\xi) = 0$ and, by the choice of M, $g(x) = 0$. Apply the MVT repeatedly:

$$0 = g(x) - g(\xi) = g'(c_1)(x-\xi) \Rightarrow g'(c_1) = 0; \quad 0 = g'(c_1) - g'(\xi) = g''(c_2)(c_1 - \xi) \Rightarrow g''(c_2) = 0; \ \dots$$

up to $g^{(n+1)}(c_{n+1}) = 0$, i.e. $f^{(n+1)}(c_{n+1}) - (n+1)!M = 0$, so $M = \dfrac{f^{(n+1)}(c_{n+1})}{(n+1)!}$. ∎

Remarks.

1. A standard way to compute f is to expand it in a Taylor/Maclaurin series.
2. If $f(\xi)$ is known and we need $f(\xi + h)$ for a small correction h:

$$f(\xi+h) = f(\xi) + f'(\xi)h + \frac{f''(\xi)}{2!}h^2 + \cdots + \frac{f^{(n)}(\xi)}{n!}h^n + R_n(h), \quad R_n(h) = \frac{f^{(n+1)}(\xi+\theta h)}{(n+1)!}h^{n+1},\ 0 < \theta < 1.$$

3.3.3 Example: $\sqrt{10}$ by Taylor Series about 9

$\sqrt{10} = \sqrt{3^2+1}$, so take $f(x) = x^{1/2}$, $\xi = 9$, $h = 1$:

$$\begin{aligned} f'(x) &= \tfrac{1}{2}x^{-1/2}, & f'(9) &= \tfrac{1}{6}, \\ f''(x) &= -\tfrac{1}{2^2}x^{-3/2}, & f''(9) &= -\tfrac{1}{108}, \\ f'''(x) &= \tfrac{3}{2^3}x^{-5/2}, & f'''(9) &= \tfrac{1}{648}. \end{aligned}$$

$$\sqrt{10} = 3 + \frac{1}{6} - \frac{1}{2\cdot 108} + \frac{1}{6\cdot 648} + R_3 = 3.16229 + R_3 \approx 3.162 + R_3,$$

$$|R_3| = \left|\frac{f^{(4)}(9+\theta)}{4!}\right| = \frac{1}{4!}\cdot\frac{3\cdot 5}{2^4}(9+\theta)^{-7/2} < \frac{15}{384}\cdot 9^{-7/2} \approx 0.0000179.$$

▶ *True value* $\sqrt{10} = 3.1622777$. *(The slides quote the bound as* 0.0000123*; either way three decimals are guaranteed.)*

3.3.4 Example: $\sqrt{10}$ by Maclaurin Series

$\sqrt{10} = 3\left(1+\frac{1}{9}\right)^{1/2}$; take $f(x) = (1+x)^{1/2}$, $x = \frac{1}{9}$:

$$f'(0) = \tfrac{1}{2}, \quad f''(0) = -\tfrac{1}{4}, \quad f'''(0) = \tfrac{3}{8},$$

$$\left(1+\tfrac{1}{9}\right)^{1/2} = 1 + \frac{1}{2}\cdot\frac{1}{9} - \frac{1}{8}\cdot\frac{1}{81} + \frac{1}{16}\cdot\frac{1}{729} + R_3 = 1.05410 + R_3,$$

$$R_3 = \frac{f^{(4)}(\theta/9)}{4!}\left(\frac{1}{9}\right)^4 = \frac{1}{4!}\cdot\frac{-15}{16}\left(1+\frac{\theta}{9}\right)^{-7/2}\frac{1}{9^4}, \qquad |R_3| < \frac{15}{2519424} < 0.000006.$$

Hence $\sqrt{10} \approx 3 \times 1.05410 = 3.1623$.

3.4 The Exponential Function

$$e^x = 1 + x + \frac{x^2}{2!} + \cdots + \frac{x^n}{n!} + \cdots, \qquad -\infty < x < \infty, \qquad R_n(x) = \frac{e^{\theta x}}{(n+1)!}x^{n+1} \ (0 < \theta < 1).$$

For large $|x|$ the error of $P_n(x)$ is intolerable (and many terms are needed), so split x:

$$x = E(x) + q, \qquad E(x) = \text{integer part}, \quad 0 \le q < 1, \qquad e^x = e^{E(x)} \cdot e^q.$$

- $e^{E(x)}$ by repeated multiplication: $e \cdot e \cdots e$ ($E(x)$ times) if $E(x) \ge 0$, or $\frac{1}{e} \cdots \frac{1}{e}$ ($-E(x)$ times) if $E(x) < 0$, with $e = 2.718281828459045\ldots$ and $\frac{1}{e} = 0.367879441171442\ldots$
- $e^q = \sum_{n\ge 0} \frac{q^n}{n!}$ with a small argument.

3.4.1 A Sharper Remainder Bound

The crude bound is $0 \le R_n(q) = \frac{e^{\theta q}}{(n+1)!}q^{n+1} < \frac{3}{(n+1)!}q^{n+1}$. Better:

$$\begin{aligned} R_n(q) &= \frac{q^{n+1}}{(n+1)!}\left[1 + \frac{q}{n+2} + \frac{q^2}{(n+2)(n+3)} + \cdots\right] \\ &< \frac{q^{n+1}}{(n+1)!}\left[1 + \frac{q}{n+2} + \left(\frac{q}{n+2}\right)^2 + \cdots\right] = \frac{q^{n+1}}{(n+1)!} \cdot \frac{1}{1 - \frac{q}{n+2}}, \end{aligned}$$

using the geometric series $a + ar + ar^2 + \cdots = \frac{a}{1-r}$, $|r| < 1$. Since $\dfrac{n+2}{(n+1)(n+2-q)} \le \dfrac{n+2}{(n+1)^2} < \dfrac{1}{n}$ for $0 \le q < 1$,

$$\boxed{0 < R_n(q) < \frac{q^n}{n!} \cdot \frac{q}{n} = u_n \frac{q}{n}}, \qquad u_k = \frac{q^k}{k!}.$$

Recurrence (one multiplication + one division per term):

$$u_0 = 1, \qquad u_k = u_{k-1} \cdot \frac{q}{k}, \qquad e^q = u_0 + u_1 + \cdots + u_n + R_n(q).$$

Stopping rule. For $n \ge 2\,|q|$,

$$|R_n(q)| < \frac{|q|^{n+1}}{(n+1)!} \cdot \frac{1}{1 - \frac{|q|}{n+2}} < \frac{2\,|q|^{n+1}}{(n+1)!} = \frac{2\,|q|}{n+1} \cdot \frac{|q|^n}{n!} < |u_n|\,.$$

▶ *So it suffices to stop as soon as $|u_n| \le \epsilon$: the first neglected-tail is smaller than the last term kept.*

3.4.2 Example: $\sqrt{e} = e^{1/2}$ within 10^{-5}

$q = \frac{1}{2}$, $u_k = u_{k-1} \cdot \dfrac{1}{2k}$:

$$\begin{aligned} &u_0 = 1, && u_1 = u_0/2 = 0.5000000, \\ &u_2 = u_1/4 = 0.1250000, && u_3 = u_2/6 = 0.0208333, \\ &u_4 = u_3/8 = 0.0026042, && u_5 = u_4/10 = 0.0002604, \\ &u_6 = u_5/12 = 0.0000217, && u_7 = u_6/14 = 0.0000016. \end{aligned}$$

$S_7 = 1.6487212$, so $\sqrt{e} = 1.64872$ (true: 1.6487213).

3.5 Sine and Cosine

By periodicity and symmetry it suffices to compute on $0 \le x \le \frac{\pi}{4}$:

$$\sin x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}, \qquad \cos x = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(2k)!}.$$

If $\frac{\pi}{4} \le x \le \frac{\pi}{2}$, use $z = \frac{\pi}{2} - x \in [0, \frac{\pi}{4}]$ and $\sin x = \cos z$ (similarly $\cos x = \sin z$).

Recurrences:

$$\sin x = \sum_{k=1}^{n} u_k + R_n(x), \qquad u_1 = x, \quad u_{k+1} = -\frac{x^2}{2k(2k+1)}\, u_k;$$

$$\cos x = \sum_{k=1}^{n} v_k + R_n(x), \qquad v_1 = 1, \quad v_{k+1} = -\frac{x^2}{(2k-1)2k}\, v_k.$$

Error. Both series are alternating with decreasing terms (for $0 \le x \le \frac{\pi}{4}$), so the error is at most the first omitted term:

$$|R_n(x)| \le \frac{|x|^{2n+1}}{(2n+1)!} = |u_{n+1}| \ \text{(sin)}, \qquad |R_n(x)| \le |v_{n+1}| \ \text{(cos)}.$$

3.5.1 Example: $\sin(23°54')$ within 10^{-4}

Convert to radians: $23°54' = 23 + \frac{54}{60} = 23.9°$ (in general degree $+\frac{\text{min}}{60} + \frac{\text{sec}}{3600}$), $x = \frac{23.9}{180}\pi = 0.41713$.

$$u_1 = 0.41713, \quad u_2 = -\frac{x^2 u_1}{2\cdot 3} = -0.01210, \quad u_3 = -\frac{x^2 u_2}{4\cdot 5} = +0.00011, \quad u_4 = -\frac{x^2 u_3}{6\cdot 7} = -0.000002.$$

$|u_4| < 10^{-4}$, so $\sin(23°54') \approx 0.41713 - 0.01210 + 0.00011 = 0.4051$ (true: 0.405142).

3.6 Hyperbolic Sine and Cosine

$$\sinh x = \frac{e^x - e^{-x}}{2} = x + \frac{x^3}{3!} + \frac{x^5}{5!} + \cdots, \qquad \cosh x = \frac{e^x + e^{-x}}{2} = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \cdots$$

(all terms positive, so no alternating-series bound).

Recurrences:

$$u_1 = x, \quad u_{k+1} = \frac{x^2}{2k(2k+1)} u_k \ \text{(sinh)}; \qquad v_1 = 1, \quad v_{k+1} = \frac{x^2}{(2k-1)2k} v_k \ \text{(cosh)}.$$

Error for sinh**.** Since $\sinh(-x) = -\sinh x$ assume $x > 0$, and let $0 \le x \le n$:

$$\begin{aligned} R_n(x) &= \frac{x^{2n+1}}{(2n+1)!} + \frac{x^{2n+3}}{(2n+3)!} + \cdots \\ &< \frac{x^{2n+1}}{(2n+1)!}\left[1 + \frac{x^2}{(2n+2)(2n+3)} + \frac{x^4}{(2n+2)^2(2n+3)^2} + \cdots\right] = \frac{x^{2n+1}}{(2n+1)!} \cdot \frac{1}{1 - \frac{x^2}{(2n+2)(2n+3)}}. \end{aligned}$$

With $x \le n$: $\dfrac{x^2}{(2n+2)(2n+3)} \le \dfrac{n^2}{(2n+2)(2n+3)} \le \dfrac{1}{4}$, so the factor is $\le \frac{4}{3}$, and

$$R_n < \frac{4}{3} u_{n+1} = \frac{4}{3} \cdot \frac{x^2}{2n(2n+1)} u_n < \frac{4}{3} \cdot \frac{1}{4} u_n \implies \boxed{R_n(x) < \tfrac{1}{3} u_n}.$$

Error for cosh**.** Similarly (assume $x > 0$, $x \le n$):

$$R_n < \frac{4}{3} v_{n+1} = \frac{4}{3} \cdot \frac{x^2}{(2n-1)2n} v_n \le \frac{4}{3} \cdot \frac{n^2}{(2n-1)2n} v_n \le \frac{4}{3} \cdot \frac{1}{2} v_n \implies \boxed{R_n(x) < \tfrac{2}{3} v_n},$$

using $x \le n$ and $2n - 1 \ge n$ (so $\frac{n^2}{(2n-1)2n} = \frac{n}{2(2n-1)} \le \frac{1}{2}$). The factor $\frac{4}{3}$ comes from the same geometric-series argument, since $\frac{x^2}{(2n+1)(2n+2)} \le \frac{1}{4}$.

3.6.1 Example: $\sinh 1.4$ within 10^{-5}

$x^2 = 1.96$:

$$\begin{aligned} u_1 &= 1.4, & u_2 &= \frac{x^2 u_1}{2\cdot 3} = 0.4573333, \\ u_3 &= \frac{x^2 u_2}{4\cdot 5} = 0.0448187, & u_4 &= \frac{x^2 u_3}{6\cdot 7} = 0.0020915, \\ u_5 &= \frac{x^2 u_4}{8\cdot 9} = 0.0000569, & u_6 &= \frac{x^2 u_5}{10\cdot 11} = 0.0000010. \end{aligned}$$

$\frac{1}{3} u_6 < 10^{-5}$, so $\sinh 1.4 = 1.904301$ (true: 1.9043015).

3.7 The Natural Logarithm

From

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots + (-1)^{n-1}\frac{x^n}{n} + \cdots \quad (-1 < x \le 1),$$

$$\ln(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \cdots,$$

subtracting gives a series with only odd powers (converges twice as fast):

$$\ln\frac{1-x}{1+x} = -2\left(x + \frac{x^3}{3} + \frac{x^5}{5} + \cdots\right).$$

Put $\dfrac{1-x}{1+x} = z$, i.e. $x = \dfrac{1-z}{1+z}$:

$$\ln z = -2\left[\frac{1-z}{1+z} + \frac{1}{3}\left(\frac{1-z}{1+z}\right)^3 + \frac{1}{5}\left(\frac{1-z}{1+z}\right)^5 + \cdots\right], \qquad 0 < z < \infty.$$

3.7.1 Range Reduction and Error Bound

For $x > 0$ write $x = 2^m z$ with integer m and $\frac{1}{2} \le z < 1$. Then $\ln x = m\ln 2 + \ln z$. Let $\xi = \dfrac{1-z}{1+z}$; then $0 < \xi \le \frac{1}{3}$ and

$$\ln x = m\ln 2 - 2\left(\xi + \frac{\xi^3}{3} + \frac{\xi^5}{5} + \cdots + \frac{\xi^{2n-1}}{2n-1}\right) - R_n(\xi).$$

The total error is the error of $m\ln 2$ plus R_n:

$$R_n(\xi) = 2\left(\frac{\xi^{2n+1}}{2n+1} + \frac{\xi^{2n+3}}{2n+3} + \cdots\right) < \frac{2\xi^{2n+1}}{2n+1}(1 + \xi^2 + \xi^4 + \cdots) = \frac{2}{1-\xi^2}\cdot\frac{\xi^{2n+1}}{2n+1}.$$

For $0 < \xi \le \frac{1}{3}$: $\dfrac{2}{1-\xi^2} \le \dfrac{9}{4}$, so

$$0 < R_n < \frac{9}{4}\cdot\frac{\xi^{2n+1}}{2n+1} \le \frac{9}{4}\cdot\frac{\xi^2\,\xi^{2n-1}}{2n-1} \le \frac{9}{4}\cdot\frac{1}{9}u_n \implies \boxed{R_n < \tfrac{1}{4}u_n}, \qquad u_k = \frac{\xi^{2k-1}}{2k-1}.$$

Summary: $\ln x = m\ln 2 - 2(u_1 + u_2 + \cdots + u_n) - R_n$, with $\ln 2 = 0.6931472\ldots$

3.7.2 Example: $\ln 3$ within 10^{-5}

$3 = 2^2 \times \frac{3}{4}$, so $m = 2$, $z = 0.75$, $\xi = \dfrac{0.25}{1.75} = \dfrac{1}{7} = 0.1428571$.

$$u_1 = \xi = 0.1428571, \quad u_2 = \frac{\xi^3}{3} = 0.0009718, \quad u_3 = \frac{\xi^5}{5} = 0.0000119, \quad u_4 = \frac{\xi^7}{7} = 0.0000002.$$

$u_1 + u_2 + u_3 + u_4 = 0.1438410$, so

$$\ln 3 = 2 \times 0.6931472 - 2 \times 0.1438410 = 1.09861.$$

▶ *The slides use* $\ln 2 = 0.6931718$ *(a typo) and get* 1.09866*. With the correct* $\ln 2 = 0.6931472$ *the result is* 1.098612*, matching the true* $\ln 3 = 1.0986123$.

3.8 Summary of Series Recurrences

Function	Reduction	Recurrence	Stop when
e^x	$x = E(x) + q$	$u_0 = 1,\ u_k = u_{k-1}\,q/k$	$\lvert u_n\rvert \le \epsilon\ (n \ge 2\,\lvert q\rvert)$
$\sin x$	$[0, \pi/4],\ \sin x = \cos(\frac{\pi}{2} - x)$	$u_1 = x,\ u_{k+1} = -\frac{x^2}{2k(2k+1)}u_k$	$\lvert u_{n+1}\rvert \le \epsilon$
$\cos x$	$[0, \pi/4]$	$v_1 = 1,\ v_{k+1} = -\frac{x^2}{(2k-1)2k}v_k$	$\lvert v_{n+1}\rvert \le \epsilon$
$\sinh x$	$x > 0$	$u_1 = x,\ u_{k+1} = \frac{x^2}{2k(2k+1)}u_k$	$\frac{1}{3}u_n \le \epsilon$
$\cosh x$	$x > 0$	$v_1 = 1,\ v_{k+1} = \frac{x^2}{(2k-1)2k}v_k$	$\frac{2}{3}v_n \le \epsilon$
$\ln x$	$x = 2^m z,\ \xi = \frac{1-z}{1+z}$	$u_k = \frac{\xi^{2k-1}}{2k-1}$	$\frac{1}{4}u_n \le \epsilon$

4 Root-Finding Methods

4.1 Overview

Goal: compute a root α of $f(x) = 0$. Methods covered: **Bisection**, **False Position**, **Newton–Raphson**, **Secant**, **Fixed Point**.

4.1.1 Plotting to Find Roots

Plot f and read off its intersections with the x-axis.

- $f(x) = x^4 + 2x^3 - x - 1$: roots $x \approx 0.86$ and $x \approx -1.86$.
- $f(x) = \sin x + x - 1$: root $x \approx 0.5$.

Problem: plotting is often hard or imprecise. Solution: numerical methods — but plotting is still the best way to get a good *initial guess*.

4.1.2 General Structure of an Iterative Method

1. **Initial guess:** a point x_0 or an interval $[x_0, x_1]$.
2. **Iteration:** produce the sequence $x_0, x_1, x_2, \ldots$
3. **Analysis:** the rate of convergence of the sequence.

4.1.3 Rate (Degree) of Convergence

Let $x_n \to \alpha$ and $e_n = |\alpha - x_n|$. If

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n^p} = \lambda \qquad (\lambda > 0 \text{ constant}),$$

then p is the **degree (order) of convergence**. $p = 1$ (with $0 < \lambda < 1$) is *linear*, $p = 2$ is *quadratic*.

▶ *Intuition: with $p = 2$ the number of correct digits roughly* doubles *each step; with $p = 1$ a fixed number of digits is gained per step.*

4.2 Bolzano's Theorem and the IVT

Bolzano's Theorem (1817). If f is continuous on $[a, b]$ and $f(x_0) < 0 < f(x_1)$ for some $x_0, x_1 \in [a, b]$, then f is zero at some point between x_0 and x_1. Equivalently: $f(a)f(b) < 0 \Rightarrow$ at least one root in (a, b).

Intermediate Value Theorem. If f is continuous on $[a, b]$ and $f(a) \neq f(b)$, then for every γ between $f(a)$ and $f(b)$ there is $a < c < b$ with $f(c) = \gamma$. (Bolzano is the case $\gamma = 0$.)

▶ *Bolzano guarantees* at least one *root; there may be several (an odd number, counted with multiplicity). A root of even multiplicity (e.g. $x^2 = 0$) produces no sign change and cannot be bracketed.*

4.3 Bisection Method

Requirements: f real and continuous on $[a, b]$, $f(a) \cdot f(b) < 0$.

Algorithm. Let $x_0 = a$, $x_1 = b$, $x_2 = \dfrac{x_0 + x_1}{2}$. Then exactly one holds:

- $f(x_2) = 0$: x_2 is a root;
- $f(x_0)f(x_2) < 0$: a root lies in (x_0, x_2);
- $f(x_1)f(x_2) < 0$: a root lies in (x_2, x_1).

Keep the half containing the sign change and repeat.

Flowchart:

```
def bisection(f, a, b, eps):
    assert f(a) * f(b) < 0       # Bolzano condition
    while b - a >= eps:
        m = (a + b) / 2
        s = f(a) * f(m)
        if s == 0: return m       # exact root
```

```
        if s < 0:  b = m          # root in [a, m]
        else:      a = m          # root in [m, b]
    return (a + b) / 2            # or a, or b
```

Error and number of iterations (not on the slides, but standard and useful). After n halvings the bracket has length $\dfrac{b-a}{2^n}$, so the midpoint is within $\dfrac{b-a}{2^{n+1}}$ of a root. To guarantee $|x-\alpha| < \varepsilon$ with the final midpoint we need

$$n \;\geq\; \log_2 \frac{b-a}{\varepsilon} - 1.$$

Convergence is linear with $\lambda = \frac{1}{2}$: one binary digit per step, about 3.3 steps per decimal digit.

4.3.1 Example: $f(x) = 0.25e^{2x} - 1.5$

Take $x_0 = 1$ ($f = +0.347$) and $x_1 = -0.2$ ($f = -1.332$):

step	midpoint	f	new bracket
x_2	0.4	-0.944	$[0.4,\ 1]$
x_3	0.7	-0.486	$[0.7,\ 1]$
x_4	0.85	-0.132	$[0.85,\ 1]$
x_5	0.925	$+0.090$	$[0.85,\ 0.925]$

Exact root: $\alpha = \frac{1}{2}\ln 6 = 0.89588$.

4.3.2 Advantages and Disadvantages

Advantages:

- Always convergent (when the requirements hold).
- Very simple.
- Only the *sign* of f is needed, not its exact value.
- The number of iterations does not depend on f (only on $b-a$ and ε).

Disadvantage: very slow convergence.

▶ *Typical use: get a rough initial estimate, then switch to a fast method such as Newton–Raphson, which needs a good starting point.*

4.4 False Position (Regula Falsi) Method

Requirements: same as bisection — f continuous on $[x_l, x_u]$ and $f(x_l)f(x_u) < 0$.

Idea: instead of the midpoint, join $(x_l, f(x_l))$ and $(x_u, f(x_u))$ with a straight line and take the line's x-intercept:

$$\frac{0 - f(x_u)}{x - x_u} = \frac{f(x_u) - f(x_l)}{x_u - x_l} \quad\Longrightarrow\quad x = x_u - f(x_u)\frac{x_u - x_l}{f(x_u) - f(x_l)} = \frac{x_l f(x_u) - x_u f(x_l)}{f(x_u) - f(x_l)}.$$

Then, as in bisection, replace the endpoint whose f has the same sign as $f(x)$, so the root stays bracketed.

General step (slides' notation): after $x_0, \ldots, x_n$,

$$x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})}\, f(x_n),$$

where x_{n-1} here denotes the *other endpoint of the current bracket*. **Termination:** $|x_{n+1} - x_n| < \varepsilon$.

▶ *Why does it converge? The root always stays inside the bracket, which can only shrink. Often one endpoint gets "stuck" (all new points fall on one side of the root), which makes the method slower than expected.*

4.4.1 Example: $2x - \ln x = 7$

$f(x) = 2x - \ln x - 7$; $f(4) = -0.38629 = f(x_l)$, $f(4.5) = 0.49592 = f(x_u)$. The first chord gives

$$x_2 = 4.5 - 0.49592 \cdot \frac{0.5}{0.49592 + 0.38629} = 4.5 - 0.28107 = 4.21893.$$

n	x_n	$f(x_n)$
2	4.218934	-1.71×10^{-3}
3	4.219902	-7.30×10^{-6}
4	4.219906	-3.1×10^{-8}

Root $\alpha = 4.21991$.

▶ *The slides show* $0.88221(x - 4.5) = -0.24796$*, i.e.* $x = 4.281$*; that has two factor-of-2 slips (slope is* 1.76442*, not* 0.88221*). The correct first iterate is* 4.21893*.*

4.4.2 Rate of Convergence (Slides' Derivation)

Let $x_n = \alpha + e_n$. From the iteration,

$$e_{n+1} = e_n - \frac{(e_n - e_{n-1})f(\alpha + e_n)}{f(\alpha + e_n) - f(\alpha + e_{n-1})} = \frac{e_{n-1}f(\alpha + e_n) - e_n f(\alpha + e_{n-1})}{f(\alpha + e_n) - f(\alpha + e_{n-1})}.$$

Taylor-expand $f(\alpha + h) = f(\alpha) + hf'(\alpha) + \frac{h^2}{2!}f''(\alpha) + \cdots$ with $f(\alpha) = 0$:

$$e_{n+1} = \frac{e_{n-1}\left[e_n f' + \frac{e_n^2}{2}f'' + \cdots\right] - e_n\left[e_{n-1}f' + \frac{e_{n-1}^2}{2}f'' + \cdots\right]}{\left[e_n f' + \frac{e_n^2}{2}f'' + \cdots\right] - \left[e_{n-1}f' + \frac{e_{n-1}^2}{2}f'' + \cdots\right]} = \frac{\frac{e_{n-1}e_n}{2}f''(\alpha) + \cdots}{f'(\alpha) + \frac{e_n + e_{n-1}}{2}f''(\alpha) + \cdots}.$$

Neglecting higher powers:

$$e_{n+1} = e_n e_{n-1}\frac{f''(\alpha)}{2f'(\alpha)}. \tag{2}$$

Assume $e_{n+1} = ce_n^p$ (3), hence $e_n = ce_{n-1}^p$, i.e. $e_{n-1} = c^{-1/p}e_n^{1/p}$ (4). Substituting (3),(4) into (2):

$$ce_n^p = e_n^{1+\frac{1}{p}}\, c^{-\frac{1}{p}}\frac{f''(\alpha)}{2f'(\alpha)} \implies p = 1 + \frac{1}{p}, \qquad c^{1+\frac{1}{p}} = \frac{f''(\alpha)}{2f'(\alpha)}.$$

$p^2 - p - 1 = 0$ gives

$$p = \frac{1 + \sqrt{5}}{2} \approx 1.618 \quad \text{(the golden ratio)}.$$

▶ *Exam note: this derivation assumes the two* most recent *iterates are used, which is exactly the Secant method. Classical regula falsi keeps a fixed endpoint once it gets stuck, and then converges only* linearly*. The slides attribute* $p = 1.618$ *to false position; know both statements.*

4.4.3 Advantages and Disadvantages

Advantages: always converges; usually faster than bisection.
Disadvantages: more operations per step than bisection; if all (or most) iterates lie on one side of the root, convergence is slow.

4.5 Newton–Raphson Method

Isaac Newton (1642–1726), Joseph Raphson (1648–1715).

4.5.1 Derivation

Let x_0 approximate a root and f be differentiable near x_0. By Taylor,

$$f(x) = f(x_0) + (x - x_0)f'(x_0) + \frac{(x - x_0)^2}{2!}f''(\gamma), \qquad \gamma \text{ between } x_0, x.$$

At the root α: $0 = f(x_0) + (\alpha - x_0)f'(x_0) + \frac{(\alpha-x_0)^2}{2!}f''(\gamma)$. If $f'(x_0) \neq 0$,

$$-\frac{f(x_0)}{f'(x_0)} = (\alpha - x_0) + \frac{(\alpha - x_0)^2}{2!}\frac{f''(\gamma)}{f'(x_0)}.$$

If x_0 is close to α, the last term is negligible, so $\alpha \approx x_0 - \dfrac{f(x_0)}{f'(x_0)}$.

Iteration:

$$\boxed{x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}} \qquad \text{Termination: } |x_n - x_{n-1}| < \varepsilon.$$

Geometry: x_{n+1} is where the *tangent* line at $(x_n, f(x_n))$ crosses the x-axis.

4.5.2 Quadratic Convergence

Substituting $x_n = \alpha + e_n$:

$$e_{n+1} = e_n - \frac{f(\alpha + e_n)}{f'(\alpha + e_n)} = e_n - \frac{e_n f'(\alpha) + \frac{e_n^2}{2}f''(\alpha) + \cdots}{f'(\alpha) + e_n f''(\alpha) + \cdots} = \frac{\frac{e_n^2}{2}f''(\alpha) + \cdots}{f'(\alpha) + e_n f''(\alpha) + \cdots}.$$

$$= \frac{e_n^2 f''(\alpha)}{2f'(\alpha)} \cdot \frac{1}{1 + e_n \frac{f''(\alpha)}{f'(\alpha)}} = \frac{e_n^2 f''(\alpha)}{2f'(\alpha)}\left(1 - e_n\frac{f''(\alpha)}{f'(\alpha)} + \cdots\right) = \frac{e_n^2}{2}\frac{f''(\alpha)}{f'(\alpha)} - \frac{e_n^3}{2}\left(\frac{f''(\alpha)}{f'(\alpha)}\right)^2 + \cdots$$

Hence

$$\lim_{n\to\infty} \frac{e_{n+1}}{e_n^2} = \frac{f''(\alpha)}{2f'(\alpha)}, \qquad p = 2. \qquad \blacksquare$$

▶ *This requires $f'(\alpha) \neq 0$ (a* simple *root). At a multiple root Newton's method degrades to linear convergence.*

4.5.3 Example: $x^4 - x = 10$

$f(x) = x^4 - x - 10$, $f'(x) = 4x^3 - 1$:

$$x_{n+1} = x_n - \frac{x_n^4 - x_n - 10}{4x_n^3 - 1} = \frac{3x_n^4 + 10}{4x_n^3 - 1}.$$

$x_0 = 2,\ x_1 = 1.871,\ x_2 = 1.85578,\ x_3 = 1.855585,\ x_4 = 1.85558452522.$

4.5.4 Example: $e^{-x} = \sin x$

$f(x) = e^{-x} - \sin x$, $f'(x) = -e^{-x} - \cos x$:

$$x_{n+1} = x_n + \frac{e^{-x_n} - \sin x_n}{e^{-x_n} + \cos x_n}.$$

$x_0 = 0.6,\ x_1 = 0.5885,\ x_2 = 0.58853274.$

4.5.5 Example: $\sqrt{5}$ (and $\sqrt[k]{a}$ in general)

Newton is a good method for $\sqrt[k]{a}$: solve $f(x) = x^k - a = 0$. For $\sqrt{5}$: $f(x) = x^2 - 5$, $f'(x) = 2x$,

$$x_{i+1} = x_i - \frac{x_i^2 - 5}{2x_i} = \frac{x_i^2 + 5}{2x_i} = \frac{1}{2}\left(x_i + \frac{5}{x_i}\right) \quad \text{(Heron's formula)}.$$

i	x_i
0	2
1	2.25
2	2.236111111
3	2.236067978
4	2.236067978

In general, $x_{i+1} = \dfrac{1}{k}\left[(k-1)x_i + \dfrac{a}{x_i^{k-1}}\right]$.

4.5.6 Newton Does Not Always Converge

Let $f(x) = \operatorname{sign}(x)\sqrt{|x|}$ (root at 0). Since $\sqrt{|x|} = \sqrt{x \operatorname{sign} x}$,

$$f'(x) = \frac{1}{2\sqrt{|x|}} \quad (x \neq 0), \qquad x_{n+1} = x_n - \frac{\operatorname{sign}(x_n)\sqrt{|x_n|}}{\frac{1}{2\sqrt{|x_n|}}} = x_n - 2\operatorname{sign}(x_n)\,|x_n| = x_n - 2x_n = -x_n.$$

The iterates oscillate $x_0, -x_0, x_0, \ldots$ forever (here $f'(\alpha) = \infty$).

▶ *Other classic failures: $f'(x_n) = 0$ (horizontal tangent, division by zero); starting far from the root (divergence or jumping to another root); cycles.*

4.5.7 Advantages and Disadvantages

Advantage: converges very fast (quadratically) — if it converges.
Disadvantages: no guarantee of convergence; computing f' may be hard; everything depends strongly on f and on x_0.

▶ *Choosing x_0 is crucial; it is best to estimate x_0 by plotting f (or by a few bisection steps).*

4.6 Secant Method

Motivation: Newton needs both f and f' every iteration. Replace the derivative by a difference quotient:

$$f'(x) \approx \frac{f(x) - f(x-h)}{h} \implies x_{n+1} = x_n - \frac{h\,f(x_n)}{f(x_n) - f(x_n - h)}.$$

Taking $x_n - h = x_{n-1}$:

$$\boxed{x_{n+1} = x_n - \frac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n) = \frac{x_{n-1}f(x_n) - x_n f(x_{n-1})}{f(x_n) - f(x_{n-1})}}$$

- **Initial approximations:** two points x_0, x_1 (no sign condition needed).
- **Termination:** $|x_n - x_{n-1}| < \varepsilon$.
- **Order of convergence:** $p = \dfrac{1+\sqrt{5}}{2} \approx 1.618$ (superlinear), from the derivation above.

▶ *Secant vs. false position: identical formula, but secant always uses the* two latest *points and does* ***not*** *check Bolzano's condition. It is faster but can diverge.*

Advantages: faster than bisection; Bolzano not needed; only one new f evaluation per step (no derivative).
Disadvantages: harder than bisection; may fail to converge; no guarantee if the initial guess is far from the root.

4.7 Comparison Example: $\sqrt{11}$ by All Four Methods

$f(x) = x^2 - 11$. Since $3^2 = 9 < 11 < 16 = 4^2$: $x_l = 3$, $x_u = 4$. True value $\sqrt{11} = 3.31662479$.

	Bisection		False position	
iter	x	$f(x)$	x	$f(x)$
1	3.5	1.25	3.28571429	−0.2040816
2	3.25	−0.4375	3.31372549	−0.0192234
3	3.375	0.390625	3.31635389	−0.0017969
4	3.3125	−0.02734375	3.31659949	−0.0001678
5	3.34375	0.18066406	3.31662243	−0.0000157
6	3.328125	0.07641602	3.31662457	−0.0000015

	Newton–Raphson ($x_0 = 3$)	
iter	x	$f(x)$
0	3	-2
1	3.33333333	0.1111111
2	3.31666667	0.0002778
3	3.31662479	0.0000000

	Secant			
iter	x	$f(x)$	x	$f(x)$
-1	3	-2	2	-7
0	4	5	3	-2
1	3.28571429	-0.2040816	3.4	0.56
2	3.31372549	-0.0192234	3.3125	-0.0273438
3	3.31663837	0.0000901	3.31657356	-0.0003398
4	3.31662478	0.0000000	3.31662482	0.0000002

▶ *Observations: bisection gains ∼1 bit/step and oscillates; false position keeps $x_u = 4$ fixed (all iterates below the root) so it is only linear; Newton doubles correct digits each step; secant is between the two. Note that from $(3, 4)$ the first two steps of false position and secant coincide (both use $x = 4$); from step 3 secant drops $x = 4$ and pulls ahead. In the second secant run the start $[2, 3]$ does* not *bracket the root, yet secant still converges.*

4.8 Fixed-Point Iteration

Definition. $\alpha \in [a, b]$ is a **fixed point** of g if $g(\alpha) = \alpha$.
Example: $g(x) = x^2 - 3x + 4$ has $g(2) = 2$, so 2 is a fixed point.

Method. Rewrite $f(x) = 0$ in the form $x = g(x)$ so that fixed points of g are roots of f. Then

$$x_0 \text{ given}, \qquad x_{n+1} = g(x_n).$$

Graphically: the root is the intersection of $y = x$ and $y = g(x)$; iterates form a "staircase" or "cobweb". (Equivalently, writing $f = f_1 - f_2$, the root is the intersection of $y = f_1(x)$ and $y = f_2(x)$.)

4.8.1 Convergence Theorem

Theorem. Let $g : [a, b] \to [a, b]$ be differentiable with

$$|g'(x)| \le q < 1 \qquad \text{for all } x \in [a, b].$$

Then g has **exactly one** fixed point $\xi \in [a, b]$, and for every $x_0 \in [a, b]$ the sequence $x_{n+1} = g(x_n)$ converges to ξ.
▶ *Why: by the MVT, $|x_{n+1} - \xi| = |g(x_n) - g(\xi)| = |g'(c)|\,|x_n - \xi| \le q\,|x_n - \xi|$, so $|x_n - \xi| \le q^n\,|x_0 - \xi| \to 0$. A useful a-posteriori bound: $|x_n - \xi| \le \dfrac{q}{1-q}\,|x_n - x_{n-1}|$. Convergence is linear with ratio $\approx |g'(\xi)|$; the smaller $|g'|$, the faster. Note also: Newton's method is the fixed-point iteration with $g(x) = x - f(x)/f'(x)$, for which $g'(\alpha) = 0$.*

Flowchart:

```
def fixed_point(g, x0, eps):
    x = x0
    while True:
        x_new = g(x)
        if abs(x_new - x) < eps:
            return x_new        # root is x_new
        x = x_new
```

4.8.2 Example: $3x - 2e^{-x} = 0$

$f(0) = -2 < 0$ and $f(1) = 3 - \frac{2}{e} > 0$, so there is a root in $[0, 1]$; plotting $y = e^{-x}$ and $y = \frac{3}{2}x$ shows it is unique. Take

$$g(x) = \frac{2}{3}e^{-x}, \qquad |g'(x)| = \frac{2}{3}e^{-x} \le \frac{2}{3} < 1 \text{ on } [0, 1].$$

With $x_0 = 0.5$:

n	x_n	n	x_n
1	0.40435	6	0.43299
2	0.44494	7	0.43238
3	0.42724	8	0.43264
4	0.43487	9	0.43253
5	0.43157	10	0.43258

$|x_{10} - x_9| < 0.0001$, so $x \approx 0.4326$. (Since $g' < 0$ the iterates alternate around the root.)

4.8.3 Example: Choice of g Matters — $x^2 - x - 2 = 0$

Roots: $x = 2$ and $x = -1$.

- $g(x) = x^2 - 2$: $g'(x) = 2x$, and $|g'| < 1$ only for $-0.5 < x < 0.5$, which contains neither root. **Diverges.**
- $g(x) = \sqrt{2+x}$: $g'(x) = \dfrac{1}{2\sqrt{2+x}} \le \dfrac{1}{2\sqrt{2}} < 1$ for all $x > 0$. **Converges** to 2.
- $g(x) = 1 + \dfrac{2}{x}$: $g'(x) = -\dfrac{2}{x^2}$, and $|g'(x)| < 1$ for $x > \sqrt{2}$ (slides state $x > \frac{3}{2}$). **Converges** to 2.

▶ *The slides write $g'(x) \le \frac{2}{5}$ for $\sqrt{2+x}$ and $g'(x) = \frac{-1}{x^2}$ for $1 + \frac{2}{x}$; the correct values are $\frac{1}{2\sqrt{2+x}} < \frac{1}{2\sqrt{2}} \approx 0.354$ and $-\frac{2}{x^2}$.*

4.8.4 Advantages and Disadvantages

Advantages: none in particular (the slides say "None!"), though it is simple and is the framework behind Newton's method and iterative linear/nonlinear system solvers.
Disadvantages: no guarantee of convergence; convergence conditions are hard to check; convergence is only linear (degree 1); everything depends on the choice of g and x_0.

4.9 Summary of Root-Finding Methods

Method	Start	Needs	Order p	Guaranteed?
Bisection	$[a,b],\ f(a)f(b) < 0$	sign of f	1 ($\lambda = \frac{1}{2}$)	yes
False position	$[a,b],\ f(a)f(b) < 0$	f	1.618 (slides)/1	yes
Newton–Raphson	x_0	$f,\ f'$	2	no
Secant	$x_0,\ x_1$	f	1.618	no
Fixed point	x_0	g with $\lvert g'\rvert < 1$	1	if $\lvert g'\rvert \le q < 1$

4.10 Homework Problems (Chapter 2+)

1. Solve $2x - \ln x = 7$ with all the methods above, $\varepsilon < 10^{-4}$.
2. $3x - 2e^{-x} = 0$ was solved by fixed point; solve it with the other methods, $\varepsilon < 10^{-4}$.
3. Solve $x^2 - x - 2 = 0$ by fixed point with $g(x) = \sqrt{2+x}$ and $g(x) = 1 + \frac{2}{x}$, $\varepsilon < 10^{-4}$.

5 Systems of Linear Equations

5.1 Overview

Methods for solving linear systems fall into two groups:

- **Exact (direct) methods:** a finite number of operations; if computed exactly they yield the exact solution (only round-off spoils it). Examples: Cramer's rule, Gaussian elimination and its compact/LU variants, principal elements, Khaletsky.
- **Iterative methods:** start from an initial guess and improve it each iteration. Examples: Jacobi and Gauss–Seidel (successive approximations).

Chapter topics (from the slides): Cramer's rule; the Gaussian method; the Gaussian compact method; the Crout–Doolittle modification; the method of principal elements; the scheme of Khaletsky; the method of successive approximations (Jacobi, Gauss–Seidel).

▶ *The posted slides work out Cramer, Gauss and the compact scheme in detail. The remaining methods appear in the chapter outline only, so Sections 5.6–5.8 below summarise them from the course textbook (Kopchenova & Maron) with small worked examples verified numerically.*

5.2 Notation

A system of n equations in n unknowns:

$$\begin{aligned} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= a_{1,n+1} \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= a_{2,n+1} \\ &\vdots \\ a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n &= a_{n,n+1} \end{aligned} \qquad \Longleftrightarrow \qquad A\mathbf{x} = \mathbf{b},$$

with $A = [a_{ij}]_{n\times n}$, $\mathbf{b} = (a_{1,n+1}, \ldots, a_{n,n+1})^T$, $\mathbf{x} = (x_1, \ldots, x_n)^T$. Writing the constants as column $n+1$ of the augmented matrix is the convention used in all the tableaux below.

5.3 Cramer's Rule

If A is non-singular ($\det A \neq 0$) the system has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$, and

$$x_j = \frac{\det B_j}{\det A},$$

where B_j is A with its j-th column replaced by $\mathbf{b}$.

Cost. Cramer needs $n+1$ determinants — very laborious for large n. Computing a determinant costs $\mathcal{O}(n^\omega)$ for some $\omega \geq 2$; the fastest matrix-multiplication algorithms (Coppersmith–Winograd) reach $\mathcal{O}(n^{2.376})$ but use heavy machinery and are impractical. (By cofactor expansion a determinant costs $\mathcal{O}(n!)$.)

▶ *In practice: Cramer is fine for $n = 2, 3$ by hand (we use it in Newton's method for 2 equations, Section 6), but never for large n.*

5.4 The Gaussian Method (Successive Elimination)

Carl Friedrich Gauss (1777–1855).

5.4.1 Warm-up Example (3×3, row operations)

$$\begin{cases} x_1 + 2x_2 + 3x_3 = 6 \\ 2x_1 - x_2 + 4x_3 = 8 \\ -x_1 + 8x_2 + 2x_3 = 12 \end{cases}$$

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & -1 & 4 & 8 \\ -1 & 8 & 2 & 12 \end{array}\right] \xrightarrow[R_3+R_1]{R_2-2R_1} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -5 & -2 & -4 \\ 0 & 10 & 5 & 18 \end{array}\right] \xrightarrow{R_3+2R_2} \left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 0 & -5 & -2 & -4 \\ 0 & 0 & 1 & 10 \end{array}\right]$$

Back substitution: $x_3 = 10$, $x_2 = -\frac{16}{5}$, $x_1 = -\frac{88}{5}$.

5.4.2 The General Scheme (shown for $n = 4$)

Forward procedure (elimination).

Step 1. Let $a_{11} \neq 0$ be the **leading element**. Divide equation 1 by it:

$$x_1 + b_{12}x_2 + b_{13}x_3 + b_{14}x_4 = b_{15}, \qquad b_{1j} = \frac{a_{1j}}{a_{11}} \ (j = 2, \ldots, 5).$$

Eliminate x_1 from equations $2, 3, 4$:

$$a_{ij}^{(1)} = a_{ij} - a_{i1}b_{1j}, \qquad i = 2, 3, 4;\ j = 2, \ldots, 5.$$

Step 2. Divide the first reduced equation by $a_{22}^{(1)}$:

$$x_2 + b_{23}^{(1)}x_3 + b_{24}^{(1)}x_4 = b_{25}^{(1)}, \qquad b_{2j}^{(1)} = \frac{a_{2j}^{(1)}}{a_{22}^{(1)}} \ (j = 3, 4, 5),$$

and eliminate x_2: $a_{ij}^{(2)} = a_{ij}^{(1)} - a_{i2}^{(1)}b_{2j}^{(1)}$, $i = 3, 4$; $j = 3, 4, 5$.

Step 3. $x_3 + b_{34}^{(2)}x_4 = b_{35}^{(2)}$ with $b_{3j}^{(2)} = a_{3j}^{(2)}/a_{33}^{(2)}$; eliminate x_3: $a_{4j}^{(3)} = a_{4j}^{(2)} - a_{43}^{(2)}b_{3j}^{(2)}$, $j = 4, 5$.

The result is an equivalent **triangular** system:

$$\begin{aligned} x_1 + b_{12}x_2 + b_{13}x_3 + b_{14}x_4 &= b_{15} \\ x_2 + b_{23}^{(1)}x_3 + b_{24}^{(1)}x_4 &= b_{25}^{(1)} \\ x_3 + b_{34}^{(2)}x_4 &= b_{35}^{(2)} \\ a_{44}^{(3)}x_4 &= a_{45}^{(3)} \end{aligned}$$

Reverse procedure (back substitution):

$$x_4 = \frac{a_{45}^{(3)}}{a_{44}^{(3)}}, \quad x_3 = b_{35}^{(2)} - b_{34}^{(2)}x_4, \quad x_2 = b_{25}^{(1)} - b_{24}^{(1)}x_4 - b_{23}^{(1)}x_3, \quad x_1 = b_{15} - b_{14}x_4 - b_{13}x_3 - b_{12}x_2.$$

General formulas (step $k = 1, \ldots, n$):

$$b_{kj}^{(k-1)} = \frac{a_{kj}^{(k-1)}}{a_{kk}^{(k-1)}}, \qquad a_{ij}^{(k)} = a_{ij}^{(k-1)} - a_{ik}^{(k-1)}\, b_{kj}^{(k-1)} \quad (i, j > k), \qquad x_i = b_{i,n+1}^{(i-1)} - \sum_{j=i+1}^{n} b_{ij}^{(i-1)} x_j.$$

5.4.3 Operation Count

For n equations:

Multiplications (left side)	$(n^2 - n) + \big((n-1)^2 - (n-1)\big) + \cdots + 1 = \dfrac{n^3 - n}{3}$
Multiplications (right side)	$(n-1) + (n-2) + \cdots + 1 = \dfrac{n(n-1)}{2}$
Divisions	$1 + \cdots + (n-1) + n + n = \dfrac{n(n+1)}{2} + n$
Total	$\dfrac{n^3 - n}{3} + n^2 + n \;\simeq\; \dfrac{1}{3}n^3$

▶ *Gauss is $\mathcal{O}(n^3)$, while Cramer with cofactor determinants is $\mathcal{O}(n!)$ — this is why elimination is the standard direct method.*

5.4.4 Worked Example (4×4)

$$\begin{aligned} 2.0x_1 + 1.0x_2 - 0.1x_3 + 1.0x_4 &= 2.7 \\ 0.4x_1 + 0.5x_2 + 4.0x_3 - 8.5x_4 &= 21.9 \\ 0.3x_1 - 1.0x_2 + 1.0x_3 + 5.2x_4 &= -3.9 \\ 1.0x_1 + 0.2x_2 + 2.5x_3 - 1.0x_4 &= 9.9 \end{aligned}$$

Step 1. Divide by $a_{11} = 2$: $x_1 + 0.5x_2 - 0.05x_3 + 0.5x_4 = 1.35$. For $i = 2$:

$a_{22}^{(1)} = 0.5 - 0.4(0.5) = 0.3$, $a_{23}^{(1)} = 4 + 0.4(0.05) = 4.02$, $a_{24}^{(1)} = -8.5 - 0.4(0.5) = -8.7$, $a_{25}^{(1)} = 21.9 - 0.4(1.35) = 21.36$.

Similarly for $i = 3, 4$, giving

$$\begin{aligned} 0.3x_2 + 4.02x_3 - 8.7x_4 &= 21.36 \\ -1.15x_2 + 1.015x_3 + 5.05x_4 &= -4.305 \\ -0.3x_2 + 2.55x_3 - 1.5x_4 &= 8.55 \end{aligned}$$

Step 2. Divide by $a_{22}^{(1)} = 0.3$: $x_2 + 13.40x_3 - 29.00x_4 = 71.20$. Then

$a_{33}^{(2)} = 1.015 + 1.15(13.40) = 16.425$, $a_{34}^{(2)} = 5.05 - 1.15(29.00) = -28.300$, $a_{35}^{(2)} = -4.305 + 1.15(71.20) = 77.575$,

and similarly for $i = 4$:

$$16.425x_3 - 28.300x_4 = 77.575, \qquad 6.570x_3 - 10.200x_4 = 29.910.$$

Step 3. Divide by 16.425: $x_3 - 1.72298x_4 = 4.72298$. Then

$$a_{44}^{(3)} = -10.200 + 6.570(1.72298) = 1.11998, \qquad a_{45}^{(3)} = 29.910 - 6.570(4.72298) = -1.11998.$$

Back substitution:

$$x_4 = -1.00000, \quad x_3 = 4.72298 - 1.72298 = 3.00000,$$
$$x_2 = 71.20 - 13.40(3) + 29.0 = 2.00000, \quad x_1 = 1.35 - 0.5(2) + 0.05(3) + 0.5 = 1.00000.$$

5.4.5 Check Sums

Add a control column

$$a_{i,n+2} = \sum_{j=1}^{n} a_{ij} + a_{i,n+1} \qquad \text{(row sum including the constant term).}$$

Treat it as an extra right-hand side and apply *every* operation to it too. Since

$$\sum_{j=1}^{n} a_{ij}(x_j + 1) = \sum_{j=1}^{n} a_{ij}x_j + \sum_{j=1}^{n} a_{ij} = a_{i,n+1} + \sum_{j=1}^{n} a_{ij} = a_{i,n+2},$$

the system with right-hand side $a_{i,n+2}$ has the solution $\tilde{x}_j = x_j + 1$.

- **During the forward pass:** in every new row, the transformed control entry must equal the sum of the other entries of that row (up to one unit of the last retained digit).
- **After the reverse pass:** the control solutions must satisfy $\tilde{x}_i = x_i + 1$.

5.5 The Gaussian Compact Scheme

When computing by hand (or on a desk calculator) with single division, much time is spent writing down intermediate systems. The **compact scheme** records *all* results in one table, together with check sums.

5.5.1 Structure of the Table ($n = 4$)

Columns: i, $a_{i1}, \ldots, a_{i4}$, a_{i5} (constants), $a_{i6} = \sum_{j=1}^{5} a_{ij}$ (control). Sections:

Section I. Rows 1–4: the original coefficients and their row sums. A fifth (*marked*) row: row 1 divided by a_{11}, i.e. $1, b_{12}, \ldots, b_{15}$ and $b_{16} = a_{16}/a_{11}$.
Check: $b_{16} = 1 + \sum_{j=2}^{5} b_{1j}$ to within one unit of the last digit; otherwise redo the step.

Section II. Rows 2–4: $a_{ij}^{(1)} = a_{ij} - a_{i1}b_{1j}$ $(j = 2, \ldots, 6)$, checked against row sums; then the marked row: row 2 of Section II divided by $a_{22}^{(1)}$, giving $1, b_{23}^{(1)}, b_{24}^{(1)}, b_{25}^{(1)}, b_{26}^{(1)}$.

Section III. Rows 3–4: $a_{ij}^{(2)} = a_{ij}^{(1)} - a_{i2}^{(1)}b_{2j}^{(1)}$ $(j = 3, \ldots, 6)$; marked row: divided by $a_{33}^{(2)}$.

Section IV. $a_{4j}^{(3)} = a_{4j}^{(2)} - a_{43}^{(2)}b_{3j}^{(2)}$, $j = 4, 5, 6$.

Section V (reverse). $x_4 = a_{45}^{(3)}/a_{44}^{(3)}$; then x_3, x_2, x_1 from the *marked* rows of III, II, I (the "1" entries show which coefficient multiplies which unknown). The control column is solved the same way and must give $\tilde{x}_i = x_i + 1$.

5.5.2 Example (same 4×4 system)

	i	a_{i1}	a_{i2}	a_{i3}	a_{i4}	a_{i5}	$a_{i6} = \Sigma$
I	1	2.0	1.0	−0.1	1.0	2.7	6.6
	2	0.4	0.5	4.0	−8.5	21.9	18.3
	3	0.3	−1.0	1.0	5.2	−3.9	1.6
	4	1.0	0.2	2.5	−1.0	9.9	12.6
		1	0.50	−0.05	0.50	1.35	3.30
II	2		0.30	4.02	−8.70	21.36	16.98
	3		−1.15	1.015	5.05	−4.305	0.610
	4		−0.30	2.55	−1.50	8.55	9.300
			1	13.40	−29.00	71.20	56.60
III	3			16.425	−28.300	77.575	65.700
	4			6.570	−10.200	29.910	26.280
				1	−1.72298	4.72298	4.00000
IV	4				1.11998	−1.11998	0
V					$x_4 = -1.00000$		$\tilde{x}_4 = 0.00000$
				$x_3 = 3.00000$			$\tilde{x}_3 = 4.00000$
			$x_2 = 2.00000$				$\tilde{x}_2 = 3.00000$
		$x_1 = 1.00000$					$\tilde{x}_1 = 2.00000$

Checks: e.g. $1 + 0.50 - 0.05 + 0.50 + 1.35 = 3.30$ ✓; $0.30 + 4.02 - 8.70 + 21.36 = 16.98$ ✓; and every $\tilde{x}_i = x_i + 1$ ✓.

5.6 Method of Principal Elements (Pivoting)

Problem with plain Gauss: it fails if a leading element is 0, and loses accuracy when a leading element is small (we divide by it, magnifying round-off).

Method of principal elements (complete pivoting):

1. In the matrix $[a_{ij}]$ (coefficients only) choose the element of **largest modulus**, a_{pq}: the *principal element*. Row p is the *principal row*.
2. For every other row $i \neq p$ compute the multiplier $m_i = -\dfrac{a_{iq}}{a_{pq}}$ and add $m_i\times$(principal row) to row i. This eliminates x_q.
3. Delete the principal row (store it) and column q; repeat on the reduced $(n-1)\times(n-1)$ system.
4. Finally solve the stored principal rows in reverse order (each contains one new unknown).

Variants: *partial pivoting* (largest $|a_{ik}|$ in the current column only, then swap rows) is the usual practical compromise.

▶ *Gauss's elimination is the special case with principal element $a_{11}, a_{22}^{(1)}, \ldots$ chosen by position. Choosing the largest element keeps all multipliers $|m_i| \leq 1$, which limits the growth of round-off error.*

5.6.1 Example: Why Pivoting Matters

Solve with 3 significant digits:

$$0.0001x + y = 1, \qquad x + y = 2 \qquad \text{(exact: } x = 1.0001\ldots,\ y = 0.9999\ldots\text{)}.$$

Without pivoting: $m = 10^4$; $y(1 - 10^4) = 2 - 10^4$ gives $-9999y = -9998$, rounded $-1.00 \times 10^4\, y = -1.00 \times 10^4$, so $y = 1.00$; then $x = (1 - y)/0.0001 = 0$. **Wrong.**

With pivoting (principal element 1 in row 2): subtract $0.0001\times$row 2 from row 1: $(1 - 0.0001)y = 1 - 0.0002$, so $y = 1.00$, then $x = 2 - y = 1.00$. **Correct.**

5.6.2 Example: Principal Elements on the 3×3 System

$$\left[\begin{array}{ccc|c} 1 & 2 & 3 & 6 \\ 2 & -1 & 4 & 8 \\ -1 & 8 & \mathbf{2} & 12 \end{array}\right] : \quad \text{principal element } a_{32} = 8.$$

$m_1 = -\frac{2}{8} = -\frac{1}{4}$, $m_2 = -\frac{-1}{8} = \frac{1}{8}$. Adding $m_i \times R_3$ eliminates x_2:

$$1.25x_1 + 2.5x_3 = 3, \qquad 1.875x_1 + 4.25x_3 = 9.5.$$

Principal element now 4.25 (coefficient of x_3): $m = -\frac{2.5}{4.25}$, giving $\frac{5}{34}x_1 = -\frac{44}{17}$, so $x_1 = -\frac{88}{5} = -17.6$. Back through the stored principal rows:

$$x_3 = \frac{9.5 - 1.875(-17.6)}{4.25} = 10, \qquad 8x_2 = 12 + x_1 - 2x_3 = 12 - 17.6 - 20 = -25.6 \ \Rightarrow\ x_2 = -3.2,$$

where the last line uses the first principal row $-x_1 + 8x_2 + 2x_3 = 12$. Same answer as before.

5.7 LU Factorisation: Crout–Doolittle and the Khaletsky Scheme

5.7.1 Idea

Gaussian elimination secretly factors A into a lower-triangular and an upper-triangular matrix. Once

$$A = LU,$$

solving $A\mathbf{x} = \mathbf{b}$ reduces to two triangular solves:

$$L\mathbf{y} = \mathbf{b} \ \text{(forward substitution)}, \qquad U\mathbf{x} = \mathbf{y} \ \text{(back substitution)}.$$

The factorisation costs $\sim \frac{1}{3}n^3$ once; each new right-hand side then costs only $\sim n^2$.

- **Doolittle:** L has unit diagonal ($l_{ii} = 1$).
- **Crout / Khaletsky:** U has unit diagonal ($c_{ii} = 1$). The **Khaletsky scheme** writes $A = BC$ with B lower triangular and C unit upper triangular.
- **Crout–Doolittle modification of the compact scheme:** compute each entry of the factors directly as an *accumulated sum of products* (one pass per entry), instead of storing all intermediate $a_{ij}^{(k)}$; this saves writing and, on a calculator, round-off.

5.7.2 Khaletsky (Crout) Formulas

For $A = BC$, $B = [b_{ij}]$ lower triangular, $C = [c_{ij}]$ upper triangular with $c_{ii} = 1$:

$$b_{i1} = a_{i1}, \qquad c_{1j} = \frac{a_{1j}}{b_{11}},$$

$$b_{ij} = a_{ij} - \sum_{k=1}^{j-1} b_{ik}c_{kj} \quad (i \geq j > 1), \qquad c_{ij} = \frac{1}{b_{ii}}\left(a_{ij} - \sum_{k=1}^{i-1} b_{ik}c_{kj}\right) \quad (1 < i < j).$$

Order: alternate — column j of B, then row j of C.
Solve $B\mathbf{y} = \mathbf{b}$ and $C\mathbf{x} = \mathbf{y}$:

$$y_i = \frac{1}{b_{ii}}\left(a_{i,n+1} - \sum_{k=1}^{i-1} b_{ik}y_k\right), \qquad x_n = y_n, \quad x_i = y_i - \sum_{k=i+1}^{n} c_{ik}x_k.$$

▶ *Applying the same formulas to the $(n+1)$-st column treats $\mathbf{y}$ as an extra column of C; adding the control column $n+2$ gives the usual check sums.*

5.7.3 Example: Khaletsky on the 4×4 System

The factors are exactly the numbers from the Gauss example: the columns of B are the pivot columns, and C holds the b-coefficients.

$$B = \begin{bmatrix} 2 & 0 & 0 & 0 \\ 0.4 & 0.3 & 0 & 0 \\ 0.3 & -1.15 & 16.425 & 0 \\ 1.0 & -0.3 & 6.57 & 1.11998 \end{bmatrix}, \qquad C = \begin{bmatrix} 1 & 0.5 & -0.05 & 0.5 \\ 0 & 1 & 13.4 & -29 \\ 0 & 0 & 1 & -1.72298 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

For instance $b_{32} = a_{32} - b_{31}c_{12} = -1.0 - 0.3(0.5) = -1.15$ and $c_{23} = \frac{a_{23} - b_{21}c_{13}}{b_{22}} = \frac{4 + 0.4(0.05)}{0.3} = 13.4$.
Forward: $\mathbf{y} = (1.35,\ 71.2,\ 4.72298,\ -1)^T$. Back: $\mathbf{x} = (1, 2, 3, -1)^T$.

5.7.4 Example: Doolittle on the 3×3 System

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & -1 & 4 \\ -1 & 8 & 2 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ -1 & -2 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 1 & 2 & 3 \\ 0 & -5 & -2 \\ 0 & 0 & 1 \end{bmatrix}}_{U}.$$

The multipliers $2, -1, -2$ of the row operations $R_2 - 2R_1$, $R_3 + R_1$, $R_3 + 2R_2$ are the entries of L.
$L\mathbf{y} = (6, 8, 12)^T$: $y = (6, -4, 10)$. $U\mathbf{x} = \mathbf{y}$: $x_3 = 10$, $x_2 = -\frac{16}{5}$, $x_1 = -\frac{88}{5}$.

5.8 Iterative Methods: Successive Approximations

5.8.1 General Form

Rewrite $A\mathbf{x} = \mathbf{b}$ as

$$\mathbf{x} = B\mathbf{x} + \boldsymbol{\beta}$$

and iterate $\mathbf{x}^{(k+1)} = B\mathbf{x}^{(k)} + \boldsymbol{\beta}$ from some $\mathbf{x}^{(0)}$ (often $\mathbf{x}^{(0)} = \boldsymbol{\beta}$ or $\mathbf{0}$). This is the fixed-point method of Section 4 in vector form.

Sufficient convergence condition. If some matrix norm satisfies $\|B\| < 1$, e.g.

$$\|B\|_\infty = \max_i \sum_j |b_{ij}| < 1 \quad \text{or} \quad \|B\|_1 = \max_j \sum_i |b_{ij}| < 1,$$

then the iteration converges for every $\mathbf{x}^{(0)}$ to the unique solution, and

$$\|\mathbf{x} - \mathbf{x}^{(k)}\| \le \frac{\|B\|}{1 - \|B\|} \, \|\mathbf{x}^{(k)} - \mathbf{x}^{(k-1)}\|.$$

(Necessary and sufficient: spectral radius $\rho(B) < 1$.)

5.8.2 Jacobi's Method

Solve equation i for x_i (requires $a_{ii} \neq 0$):

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(a_{i,n+1} - \sum_{j \neq i} a_{ij} x_j^{(k)} \right), \qquad i = 1, \dots, n.$$

So $b_{ij} = -a_{ij}/a_{ii}$ $(i \neq j)$, $b_{ii} = 0$, $\beta_i = a_{i,n+1}/a_{ii}$.

Condition: if A is **strictly diagonally dominant**, $|a_{ii}| > \sum_{j \neq i} |a_{ij}|$ for all i, then $\|B\|_\infty < 1$ and Jacobi converges.

▶ *Before iterating, reorder equations / take linear combinations so that the diagonal dominates.*

5.8.3 Gauss–Seidel Method

Same as Jacobi, but each newly computed component is used *immediately*:

$$x_i^{(k+1)} = \frac{1}{a_{ii}} \left(a_{i,n+1} - \sum_{j<i} a_{ij} x_j^{(k+1)} - \sum_{j>i} a_{ij} x_j^{(k)} \right).$$

Diagonal dominance also guarantees convergence of Gauss–Seidel, which usually converges faster than Jacobi and needs only one vector of storage.

5.8.4 Example: Jacobi vs. Gauss–Seidel

$$\begin{aligned} 10x_1 + x_2 + x_3 &= 12 \\ 2x_1 + 10x_2 + x_3 &= 13 \qquad \text{(exact: } x_1 = x_2 = x_3 = 1\text{)}. \\ 2x_1 + 2x_2 + 10x_3 &= 14 \end{aligned}$$

Strictly diagonally dominant; $\|B\|_\infty = \max(0.2, 0.3, 0.4) = 0.4 < 1$.

$$\text{Jacobi:} \quad x_1 = 1.2 - 0.1x_2 - 0.1x_3, \quad x_2 = 1.3 - 0.2x_1 - 0.1x_3, \quad x_3 = 1.4 - 0.2x_1 - 0.2x_2.$$

From $\mathbf{x}^{(0)} = \mathbf{0}$:

	Jacobi			**Gauss–Seidel**		
k	x_1	x_2	x_3	x_1	x_2	x_3
1	1.2000	1.3000	1.4000	1.20000	1.06000	0.94800
2	0.9300	0.9200	0.9000	0.99920	1.00536	0.99909
3	1.0180	1.0240	1.0300	0.99956	1.00018	1.00005
4	0.9946	0.9934	0.9916	0.99998	1.00000	1.00000
5	1.0015	1.0019	1.0024			
6	0.9996	0.9995	0.9993			

▶ *Gauss–Seidel reaches 4-digit accuracy in 4 iterations; Jacobi needs about 6–7.*

5.9 Summary of Linear-System Methods

Method	Type	Key fact
Cramer	direct	$n+1$ determinants; only for tiny n
Gauss	direct	$\frac{1}{3}n^3$ ops; forward + back substitution
Gauss compact	direct	one table + check-sum column Σ
Principal elements	direct	pivot on largest $\lvert a_{pq}\rvert$; stable
Crout–Doolittle / Khaletsky	direct	$A = LU$ $(= BC)$; cheap for many RHS
Jacobi	iterative	converges if $\lVert B\rVert < 1$ (diag. dominant)
Gauss–Seidel	iterative	uses new values at once; usually faster

6 Systems of Nonlinear Equations

6.1 Overview

One way to solve a nonlinear system is to approximate it, step by step, by a *linear* system. The principal tool of Section 4 was Newton's method; here it is extended to systems. As in one variable, Newton for systems is costly (Jacobian + linear solve each step), which motivates cheaper secant-like modifications.

Chapter topics:

- Newton's method for a system of two equations (with Cramer's rule);
- Newton's method for n equations in n unknowns;
- the method of simple iteration (contraction mapping theorem) for 2 and for n equations.

6.2 Newton's Method for Two Equations

6.2.1 Derivation

Given

$$\begin{cases} f(x_1, x_2) = 0 \\ g(x_1, x_2) = 0 \end{cases}$$

suppose $(x_1^{(p)}, x_2^{(p)})$ is the p-th approximation of an isolated root (α, β), and write

$$(\alpha, \beta) = \left(x_1^{(p)} + e_1^{(p)},\ x_2^{(p)} + e_2^{(p)}\right),$$

where $e^{(p)} = (e_1^{(p)}, e_2^{(p)})$ is the error. By Taylor's theorem (keeping linear terms):

$$0 = f(\alpha, \beta) \approx f(x_1^{(p)}, x_2^{(p)}) + e_1^{(p)} \frac{\partial f}{\partial x_1} + e_2^{(p)} \frac{\partial f}{\partial x_2}, \qquad 0 \approx g(x_1^{(p)}, x_2^{(p)}) + e_1^{(p)} \frac{\partial g}{\partial x_1} + e_2^{(p)} \frac{\partial g}{\partial x_2},$$

all derivatives evaluated at $(x_1^{(p)}, x_2^{(p)})$. In matrix form:

$$-\begin{bmatrix} f \\ g \end{bmatrix} = J \begin{bmatrix} e_1^{(p)} \\ e_2^{(p)} \end{bmatrix}, \qquad J(x_1, x_2) = \begin{bmatrix} \dfrac{\partial f}{\partial x_1} & \dfrac{\partial f}{\partial x_2} \\ \dfrac{\partial g}{\partial x_1} & \dfrac{\partial g}{\partial x_2} \end{bmatrix} \quad \textbf{(Jacobi matrix)}.$$

6.2.2 Solving the 2×2 System by Cramer's Rule

If

$$\det J = \frac{\partial f}{\partial x_1}\frac{\partial g}{\partial x_2} - \frac{\partial f}{\partial x_2}\frac{\partial g}{\partial x_1} \neq 0,$$

then

$$e_1^{(p)} = \frac{\begin{vmatrix} -f & f_{x_2} \\ -g & g_{x_2} \end{vmatrix}}{\det J}, \qquad e_2^{(p)} = \frac{\begin{vmatrix} f_{x_1} & -f \\ g_{x_1} & -g \end{vmatrix}}{\det J}.$$

Define

$$\Delta x_1^{(p)} = \begin{vmatrix} f & f_{x_2} \\ g & g_{x_2} \end{vmatrix}, \qquad \Delta x_2^{(p)} = \begin{vmatrix} f_{x_1} & f \\ g_{x_1} & g \end{vmatrix},$$

(replace the column of the unknown by $(f,g)^T$). Then the Newton update is

$$\boxed{x_1^{(p+1)} = x_1^{(p)} - \frac{\Delta x_1^{(p)}}{\det J(x_1^{(p)}, x_2^{(p)})}, \qquad x_2^{(p+1)} = x_2^{(p)} - \frac{\Delta x_2^{(p)}}{\det J(x_1^{(p)}, x_2^{(p)})}}$$

The initial approximation $(x_1^{(0)}, x_2^{(0)})$ is found roughly, usually graphically.

Summary (matrix form):

$$\begin{bmatrix} x_1^{(p+1)} \\ x_2^{(p+1)} \end{bmatrix} = \begin{bmatrix} x_1^{(p)} \\ x_2^{(p)} \end{bmatrix} - J\big(x_1^{(p)}, x_2^{(p)}\big)^{-1} \begin{bmatrix} f(x_1^{(p)}, x_2^{(p)}) \\ g(x_1^{(p)}, x_2^{(p)}) \end{bmatrix}.$$

▶ *Compare with one variable: $x_{n+1} = x_n - f(x_n)/f'(x_n)$. The derivative becomes the Jacobian, and "divide by f'" becomes "multiply by J^{-1}". Convergence is again quadratic near a root with $\det J(\alpha,\beta) \neq 0$.*

6.2.3 Example 1: $2x^3 - y^2 - 1 = 0, \quad xy^3 - y - 4 = 0$

Graphically $x_0 = 1.2$, $y_0 = 1.7$.

$$J(x,y) = \begin{bmatrix} 6x^2 & -2y \\ y^3 & 3xy^2 - 1 \end{bmatrix}, \qquad \det J(1.2, 1.7) = \begin{vmatrix} 8.64 & -3.40 \\ 4.91 & 9.40 \end{vmatrix} = 97.910.$$

$f(1.2, 1.7) = -0.434$, $g(1.2, 1.7) = 0.1956$. Hence

$$x^{(1)} = 1.2 - \frac{1}{97.910}\begin{vmatrix} -0.434 & -3.40 \\ 0.1956 & 9.40 \end{vmatrix} = 1.2 + 0.0349 = 1.2349,$$

$$y^{(1)} = 1.7 - \frac{1}{97.910}\begin{vmatrix} 8.64 & -0.434 \\ 4.91 & 0.1956 \end{vmatrix} = 1.7 - 0.0390 = 1.6610.$$

Continuing: $x^{(2)} = 1.2343$, $y^{(2)} = 1.6615$, and so on (converged value 1.23427, 1.66153).

6.2.4 Example 2: $\cos(0.4y + x^2) + x^2 + y^2 - 1.6 = 0, \quad 1.5x^2 - \frac{y^2}{0.36} - 1 = 0$

Graphically $x^{(0)} = 1.04$, $y^{(0)} = 0.47$. Partial derivatives:

$$f_x = 2x\big(1 - \sin(0.4y + x^2)\big), \quad f_y = -0.4\sin(0.4y + x^2) + 2y, \quad g_x = 3x, \quad g_y = -\frac{2y}{0.36}.$$

	$p = 0$	$p = 1$
x	1.04	1.03864
y	0.47	0.47173
f	−0.00084	0.00000
g	0.00879	0.00002
f'_x	0.09364	0.09483
f'_y	0.55801	0.56172
g'_x	3.12	3.11592
g'_y	−2.61111	−2.62072
$\det J$	−1.985549	−1.99889
Δx	−0.00271	0.00001
Δy	+0.00344	0.00000
$-\Delta x/\det J$	−0.00136	0.00000
$-\Delta y/\det J$	0.00173	0.00000

Root: $x = 1.03864$, $y = 0.47173$ (after one step the corrections vanish to 5 decimals).

6.3 Newton's Method for n Equations in n Unknowns

Given $f_j(x_1,\dots,x_n)=0$, $j=1,\dots,n$, and the p-th approximation $\mathbf{x}^{(p)}$, Taylor around $\mathbf{x}^{(p)}$ gives the linearised system

$$f_j(\mathbf{x}^{(p)})+\frac{\partial f_j}{\partial x_1}e_1^{(p)}+\cdots+\frac{\partial f_j}{\partial x_n}e_n^{(p)}=0,\qquad j=1,\dots,n.$$

With

$$J(\mathbf{x})=\begin{bmatrix}\frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n}\\ \vdots & \ddots & \vdots\\ \frac{\partial f_n}{\partial x_1} & \cdots & \frac{\partial f_n}{\partial x_n}\end{bmatrix},\qquad F(\mathbf{x})=\begin{bmatrix}f_1(\mathbf{x})\\ \vdots\\ f_n(\mathbf{x})\end{bmatrix},\ F:\mathbb{R}^n\to\mathbb{R}^n,$$

this reads $J(\mathbf{x}^{(p)})\,\mathbf{e}^{(p)}=-F(\mathbf{x}^{(p)})$, so

$$\boxed{\mathbf{x}^{(p+1)}=\mathbf{x}^{(p)}+\mathbf{e}^{(p)}=\mathbf{x}^{(p)}-J(\mathbf{x}^{(p)})^{-1}F(\mathbf{x}^{(p)})}$$

This is **Newton's method for nonlinear systems**.

▶ *Computing (and inverting) the Jacobian is the weak point of Newton's method. In practice one does not invert J: solve $J\mathbf{e}=-F$ by Gaussian elimination (Section 5). Cheaper variants freeze J for several steps or approximate it by differences (the multidimensional secant idea).*

6.3.1 Example: $x^2+y^2+z^2=1,\ 2x^2+y^2-4z=0,\ 3x^2-4y+z^2=0$

Positive solution, starting from $x_0=y_0=z_0=0.5$.

$$F(\mathbf{x})=\begin{bmatrix}x^2+y^2+z^2-1\\ 2x^2+y^2-4z\\ 3x^2-4y+z^2\end{bmatrix},\qquad J(\mathbf{x})=\begin{bmatrix}2x & 2y & 2z\\ 4x & 2y & -4\\ 6x & -4 & 2z\end{bmatrix}.$$

Iteration 1.

$$F(\mathbf{x}^{(0)})=\begin{bmatrix}-0.25\\ -1.25\\ -1.00\end{bmatrix},\quad J(\mathbf{x}^{(0)})=\begin{bmatrix}1 & 1 & 1\\ 2 & 1 & -4\\ 3 & -4 & 1\end{bmatrix},\quad \det J=-40,$$

$$J^{-1}=-\frac{1}{40}\begin{bmatrix}-15 & -5 & -5\\ -14 & -2 & 6\\ -11 & 7 & -1\end{bmatrix}=\begin{bmatrix}\frac{3}{8} & \frac{1}{8} & \frac{1}{8}\\ \frac{7}{20} & \frac{1}{20} & -\frac{3}{20}\\ \frac{11}{40} & -\frac{7}{40} & \frac{1}{40}\end{bmatrix},$$

$$\mathbf{x}^{(1)}=\begin{bmatrix}0.5\\ 0.5\\ 0.5\end{bmatrix}-J^{-1}\begin{bmatrix}-0.25\\ -1.25\\ -1.00\end{bmatrix}=\begin{bmatrix}0.5\\ 0.5\\ 0.5\end{bmatrix}-\begin{bmatrix}-0.375\\ 0\\ 0.125\end{bmatrix}=\begin{bmatrix}0.875\\ 0.500\\ 0.375\end{bmatrix}.$$

Iteration 2.

$$F(\mathbf{x}^{(1)})=\begin{bmatrix}0.15625\\ 0.28125\\ 0.43750\end{bmatrix},\quad J(\mathbf{x}^{(1)})=\begin{bmatrix}1.750 & 1 & 0.750\\ 3.500 & 1 & -4\\ 5.250 & -4 & 0.750\end{bmatrix},\quad \det J=-64.75,$$

$$J^{-1}=-\frac{1}{64.75}\begin{bmatrix}-15.25 & -3.75 & -4.75\\ -23.625 & -2.625 & 9.625\\ -19.25 & 12.25 & -1.75\end{bmatrix},\qquad \mathbf{x}^{(2)}=\mathbf{x}^{(1)}-J^{-1}F(\mathbf{x}^{(1)})=\begin{bmatrix}0.875\\ 0.500\\ 0.375\end{bmatrix}-\begin{bmatrix}0.08519\\ 0.00338\\ 0.00507\end{bmatrix}=\begin{bmatrix}0.78981\\ 0.49662\\ 0.36993\end{bmatrix}.$$

Iteration 3.

$$\mathbf{x}^{(3)}=\begin{bmatrix}0.78521\\ 0.49661\\ 0.36992\end{bmatrix},\qquad F(\mathbf{x}^{(3)})\approx\begin{bmatrix}0.00002\\ 0.00004\\ 0.00006\end{bmatrix}.$$

Stopping at the third approximation: $x=0.7852$, $y=0.4966$, $z=0.3699$.

▶ *The slides' 2nd-iteration formula writes $F(\mathbf{x}^{(0)})$; it must be $F(\mathbf{x}^{(1)})$, as used in the numbers.*

6.4 Newton as a Fixed-Point Problem

Let $G(\mathbf{x}) = \mathbf{x} - J(\mathbf{x})^{-1}F(\mathbf{x})$. Then $F(\mathbf{x}) = \mathbf{0}$ has a solution iff G has a fixed point, $G(\mathbf{x}) = \mathbf{x}$, and Newton's method is the iteration $\mathbf{x}^{(p+1)} = G(\mathbf{x}^{(p)})$.

Question: when does such a map have a (unique) fixed point that the iteration finds? Answer: the contraction mapping theorem.

6.5 Simple Iteration and the Contraction Mapping Theorem

6.5.1 Setting

A system $f_1(\mathbf{x}) = 0, \dots, f_m(\mathbf{x}) = 0$ is an equation $\mathbf{f}(\mathbf{x}) = \mathbf{0}$ with $\mathbf{f} : \mathbb{R}^n \to \mathbb{R}^m$. The linear system $A\mathbf{x} - \mathbf{b} = \mathbf{0}$ is a special case.

Number of solutions. There is no general rule:

- $x_1 + 2x_2 - 3 = 0,\ 2x_1^2 + x_2^2 - 5 = 0$ (a line and an ellipse) has **two** solutions, A and B.
- In general a system may have no solution, infinitely many, or any finite number.

We restrict to $m = n$ and rewrite the system as

$$\mathbf{x} = \mathbf{g}(\mathbf{x}),$$

which generalises both the scalar fixed-point form $x = g(x)$ (Section 4) and the linear iterative form $\mathbf{x} = B\mathbf{x} + \boldsymbol{\beta}$ (Section 5.8). Then iterate

$$\mathbf{x}_{r+1} = \mathbf{g}(\mathbf{x}_r), \qquad \mathbf{x}_0 \text{ given.}$$

6.5.2 Contraction Mapping

Definition. $\mathbf{g} : S \subseteq \mathbb{R}^n \to S$ is a **contraction mapping** on a closed region S with respect to a norm $\|\cdot\|$ if there is $L < 1$ with

$$\|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{x}')\| \le L\,\|\mathbf{x} - \mathbf{x}'\| \qquad \text{for all } \mathbf{x}, \mathbf{x}' \in S.$$

Theorem (contraction mapping). If $\mathbf{g} : S \to S$ is a contraction on a closed region $S \subseteq \mathbb{R}^n$, then

(i) $\mathbf{x} = \mathbf{g}(\mathbf{x})$ has a **unique** solution $\boldsymbol{\alpha} \in S$;

(ii) for every $\mathbf{x}_0 \in S$ the sequence $\mathbf{x}_{r+1} = \mathbf{g}(\mathbf{x}_r)$ converges to $\boldsymbol{\alpha}$.

Moreover $\|\mathbf{x}_r - \boldsymbol{\alpha}\| \le \dfrac{L^r}{1-L}\|\mathbf{x}_1 - \mathbf{x}_0\|$.

6.5.3 A Practical Test via Partial Derivatives

Let $\mathbf{g} = (g_1, \dots, g_n)^T$ with continuous $\partial g_i/\partial x_j$ on a *convex* region S. By Taylor's theorem in several variables, for $\mathbf{x}, \mathbf{x}' \in S$,

$$g_i(\mathbf{x}) - g_i(\mathbf{x}') = \frac{\partial g_i(\boldsymbol{\eta}_i)}{\partial x_1}(x_1 - x_1') + \cdots + \frac{\partial g_i(\boldsymbol{\eta}_i)}{\partial x_n}(x_n - x_n'),$$

with $\boldsymbol{\eta}_i = (1-\theta)\mathbf{x} + \theta\mathbf{x}'$ for some $\theta \in (0,1)$. Let

$$g_{ij} = \sup_{\mathbf{x} \in S} \left| \frac{\partial g_i(\mathbf{x})}{\partial x_j} \right|, \qquad G = [g_{ij}].$$

By Cauchy–Schwarz (Euclidean norm),

$$\|\mathbf{g}(\mathbf{x}) - \mathbf{g}(\mathbf{x}')\| \le \sqrt{\sum_{i,j} g_{ij}^2}\ \|\mathbf{x} - \mathbf{x}'\|.$$

▶ ***Test:*** *if $\mathbf{g}(S) \subseteq S$ and $L = \sqrt{\sum_{i,j} g_{ij}^2} < 1$, then $\mathbf{g}$ is a contraction on S. Other usable sufficient conditions (any norm works):* $\max_i \sum_j g_{ij} < 1$ *(row sums) or* $\max_j \sum_i g_{ij} < 1$ *(column sums). For two equations $x = \varphi_1(x,y)$, $y = \varphi_2(x,y)$, the textbook condition is $|\partial_x\varphi_1| + |\partial_x\varphi_2| \le q < 1$ and $|\partial_y\varphi_1| + |\partial_y\varphi_2| \le q < 1$ in the region.*

6.5.4 Example: Line and Ellipse

Solve $x_1 + 2x_2 - 3 = 0$, $2x_1^2 + x_2^2 - 5 = 0$ on $S:\ 1 \le x_1 \le 2,\ 0.5 \le x_2 \le 1.5$ (the root A). Rewrite:

$$x_1 = g_1(\mathbf{x}) = \left(\frac{5 - x_2^2}{2}\right)^{1/2}, \qquad x_2 = g_2(\mathbf{x}) = \frac{3 - x_1}{2}.$$

Derivatives:

$$\frac{\partial g_1}{\partial x_2} = -\frac{x_2}{(10 - 2x_2^2)^{1/2}},\ \text{max modulus at } x_2 = 1.5:\ \frac{3}{\sqrt{22}}; \qquad \frac{\partial g_2}{\partial x_1} = -\frac{1}{2}; \qquad \frac{\partial g_1}{\partial x_1} = \frac{\partial g_2}{\partial x_2} = 0.$$

$$G = \begin{bmatrix} 0 & \frac{3}{\sqrt{22}} \\ \frac{1}{2} & 0 \end{bmatrix}, \qquad \sqrt{\sum_{i,j} g_{ij}^2} = \sqrt{\frac{9}{22} + \frac{1}{4}} = \sqrt{\frac{29}{44}} \approx 0.81 < 1.$$

Also $\mathbf{g}(S) \subseteq S$: $g_1 \in [1.17, 1.54] \subset [1, 2]$ and $g_2 \in [0.5, 1] \subset [0.5, 1.5]$. So $\mathbf{g}$ is a contraction on S and the root is unique there.

Starting from the centre of S, (1.5, 1):

r	0	1	2	3	4	5	6
x_1	1.5	1.414	1.490	1.478	1.488	1.487	1.488
x_2	1.0	0.750	0.793	0.755	0.761	0.756	0.756

▶ *Convergence is linear with ratio about L (here ≈ 0.8 by the bound, faster in practice), compared to Newton's quadratic convergence — but no Jacobian is needed.*

6.6 Homework (Chapter 4)

Problem. Using the contraction mapping theorem, show that

$$\begin{cases} x^3 - y^3 - 6y + 2 = 0 \\ x^3 + y^3 - 6x + 3 = 0 \end{cases}$$

has a unique solution in $S:\ 0 \le x \le 1,\ 0 \le y \le 1$. Then solve it by both Newton's method and simple iteration with at least 3 correct digits.

Hints (worked check). Natural rewrite:

$$x = g_1 = \frac{x^3 + y^3 + 3}{6}, \qquad y = g_2 = \frac{x^3 - y^3 + 2}{6}.$$

- On S: $g_1 \in [\frac{1}{2}, \frac{5}{6}]$, $g_2 \in [\frac{1}{6}, \frac{1}{2}]$, so $\mathbf{g}(S) \subseteq S' = [\frac{1}{2}, \frac{5}{6}] \times [\frac{1}{6}, \frac{1}{2}] \subset S$.
- $|\partial g_i/\partial x| = \frac{x^2}{2}$, $|\partial g_i/\partial y| = \frac{y^2}{2}$. On all of S these reach $\frac{1}{2}$ and $\sqrt{\sum g_{ij}^2} = 1$ (inconclusive), but on S': $g_{i1} \le \frac{25}{72}$, $g_{i2} \le \frac{1}{8}$, so $\sqrt{\sum g_{ij}^2} = \sqrt{2\left(\frac{625}{5184} + \frac{1}{64}\right)} \approx 0.52 < 1$. Every fixed point in S lies in S', hence it is unique.
- Simple iteration from $(0.5, 0.5)$: $(0.54167, 0.33333)$, $(0.53266, 0.35365)$, $(0.53256, 0.35115)$, $(0.53239, 0.35129)$, $(0.53238, 0.35126)$, …
- Newton from $(0.5, 0.5)$: $(0.52688, 0.35484)$, $(0.53236, 0.35125)$, $(0.53237, 0.35126)$.

Solution: $x \approx 0.53237$, $y \approx 0.35126$.

7 Exam Cheat Sheet (Sections 1–4)

7.1 Errors

$\Delta_a = \lvert a - a_0 \rvert, \quad \delta_a = \Delta_a / \lvert a \rvert$	absolute / relative error
$\Delta_{a \pm b} = \Delta_a + \Delta_b$	sums: add absolute errors
$\delta_{ab} = \delta_{a/b} = \delta_a + \delta_b, \quad \delta_{x^k} = \lvert k \rvert \, \delta_x$	products: add relative errors
$\delta_r = \sqrt{3(m+n)}\, \delta$	statistical estimate, $m + n > 10$
$\Delta_y = \sum_i \lvert \partial f / \partial x_i \rvert \, \Delta_{x_i}, \quad \Delta_{\ln x} = \delta_x$	function of approximate arguments
$\Delta_{x_i} = \Delta_y / (n \lvert \partial f / \partial x_i \rvert)$	inverse problem (equal effects)

Correct digits: up to the decimal place of the first significant digit of Δ. Rounding adds the discarded part to Δ.

7.2 Values of Functions

- Horner: $b_0 = a_0$, $b_k = b_{k-1}\xi + a_k$, $P(\xi) = b_n$; $b_0, \dots, b_{n-1}$ = quotient by $(x - \xi)$. Cost: n mult., $n - k$ add.
- Taylor remainder: $R_n = \dfrac{f^{(n+1)}(c)}{(n+1)!}(x - \xi)^{n+1}$.
- $e^x = e^{E(x)}e^q$; $R_n < u_n \frac{q}{n}$ and for $n \geq 2\,|q|$ stop when $|u_n| \leq \epsilon$.
- $\sin, \cos$ on $[0, \pi/4]$; alternating, error $\leq$ first omitted term.
- sinh: $R_n < \frac{1}{3}u_n$; cosh: $R_n < \frac{2}{3}v_n$; ln: $x = 2^m z$, $\xi = \frac{1-z}{1+z} \leq \frac{1}{3}$, $R_n < \frac{1}{4}u_n$.

7.3 Root Finding

Bisection	$m = \frac{a+b}{2}$; keep sign change; $n \geq \log_2 \frac{b-a}{\varepsilon} - 1$
False position	$x = \dfrac{x_l f(x_u) - x_u f(x_l)}{f(x_u) - f(x_l)}$; keep sign change
Newton–Raphson	$x_{n+1} = x_n - \dfrac{f(x_n)}{f'(x_n)}$; $e_{n+1} \approx \dfrac{f''(\alpha)}{2f'(\alpha)} e_n^2$
Secant	$x_{n+1} = x_n - \dfrac{x_n - x_{n-1}}{f(x_n) - f(x_{n-1})} f(x_n)$; $p = \frac{1+\sqrt{5}}{2}$
Fixed point	$x_{n+1} = g(x_n)$; converges if $g : [a, b] \to [a, b]$, $\lvert g' \rvert \leq q < 1$

7.4 Linear Systems

- Cramer: $x_j = \det B_j / \det A$.
- Gauss: $b_{kj}^{(k-1)} = a_{kj}^{(k-1)} / a_{kk}^{(k-1)}$, $a_{ij}^{(k)} = a_{ij}^{(k-1)} - a_{ik}^{(k-1)} b_{kj}^{(k-1)}$; cost $\approx \frac{1}{3}n^3$.
- Check column $a_{i,n+2} = \sum_{j \leq n+1} a_{ij}$; control solution $\tilde{x}_i = x_i + 1$.
- Principal elements: pivot on the largest $|a_{pq}|$, multipliers $m_i = -a_{iq}/a_{pq}$.
- Khaletsky / Crout: $A = BC$, C unit upper; $b_{ij} = a_{ij} - \sum_{k<j} b_{ik}c_{kj}$, $c_{ij} = (a_{ij} - \sum_{k<i} b_{ik}c_{kj})/b_{ii}$.
- Jacobi / Gauss–Seidel: $x_i = \frac{1}{a_{ii}}(a_{i,n+1} - \sum_{j \neq i} a_{ij}x_j)$; converge if strictly diagonally dominant ($\|B\| < 1$).

7.5 Nonlinear Systems

- Newton (2 eqs): $x^{(p+1)} = x^{(p)} - \dfrac{\begin{vmatrix} f & f_y \\ g & g_y \end{vmatrix}}{\det J}$, $y^{(p+1)} = y^{(p)} - \dfrac{\begin{vmatrix} f_x & f \\ g_x & g \end{vmatrix}}{\det J}$.
- Newton (n eqs): $\mathbf{x}^{(p+1)} = \mathbf{x}^{(p)} - J(\mathbf{x}^{(p)})^{-1} F(\mathbf{x}^{(p)})$ (solve $J\mathbf{e} = -F$).
- Simple iteration $\mathbf{x} = \mathbf{g}(\mathbf{x})$: converges if $\mathbf{g}(S) \subseteq S$ and $\sqrt{\sum_{i,j} \sup_S |\partial g_i/\partial x_j|^2} < 1$.